

CHAPTER 1

Concepts in Writing and Learning to Write in a Second Language

DOI: [10.4324/9781003004943-1](https://doi.org/10.4324/9781003004943-1)

QUESTIONS FOR REFLECTION

1. Where do you think writing comes from, and why have writing skills become so essential in modern societies?
2. What do you think accounts for the diversity of writing systems? How do you think people best learn to use a writing system?
3. In what ways is writing in an additional language (that is, a language other than one's maternal language or languages) similar to – and different from – writing in one's mother tongue? Why? How might being literate in one language affect developing literacy in another language?
4. What sorts of knowledge about writing, writing processes, and writing pedagogy would you like to develop? Why?
5. Consider your experiences as a student writer and as a teacher of writing, if applicable. Whether you are an experienced or inexperienced teacher, what skills and competencies do you most want your students to develop? Why?

One of our primary responsibilities as teachers of language and writing is to conceptualize our beliefs and practices about learning and teaching. For some people, conceptualizing belief and practice mainly amounts to applying practical knowledge that we gain through hands-on experience. To develop deep professional expertise, though, experience alone is far from enough:

Everything we do in the classroom, the methods and materials we adopt, the teaching styles we assume, the tasks we assign, are guided by both practical and theoretical knowledge, and our decisions can be more effective if that knowledge is explicit. A familiarity with what is known about writing, and about teaching writing, can therefore help us to reflect on our assumptions and enable us to approach current teaching methods with an informed and critical eye.

([Hyland, 2019](#), p. 1)

We have revised this book with the dual aims of providing readers with explicit knowledge and practical tools for performing successfully as teachers of composition. It is customary for professional preparation manuals to present an account of a field's historical foundations, fundamental concepts, theoretical landscape, research findings, and models of practice. Pre- and

in-service teachers seeking practical strategies for effective classroom teaching occasionally find this convention to be frustrating. Newcomers to a profession may understandably find the careful study of abstract theories and research-based findings to be only distantly relevant to the day-to-day needs of future practitioners. This book aims to provide readers with a principled set of instructional tools for teaching writing to secondary and postsecondary learners of English as a second language (ESL), English as a foreign language (EFL), and English as an additional language (EAL).¹ To accomplish this objective, this chapter lays a foundation of historical precedents, theoretical principles, and research-based models so that readers can build sufficient knowledge to make informed decisions about the methods, processes, and strategies introduced in the practice-oriented chapters that follow.

By acquainting themselves with the historical origins of writing, the disciplines devoted to its study, and the philosophical orientations that guide L1 and L2 composition instruction, readers can approach teaching practice and innovation from a knowledgeable point of reference. This background knowledge helps teachers make informed instructional decisions that best serve their students in their local educational contexts ([Atkinson, 2010](#); [Freeman et al., 2019](#); [Hedgcock, 2010](#)). In this chapter, we first examine the intellectual and practical value of theoretical knowledge in composition instruction. Following a brief overview of the functions of writing in contemporary society, we examine the complex factors that distinguish multilingual writers and writing processes from monolingual writers and writing processes. To complete this overview and to lay a conceptual foundation for subsequent chapters, we survey the disciplinary sources that have shaped L2 writing as a field of study and professional specialization.

THEORETICAL KNOWLEDGE IN L2 WRITING INSTRUCTION AND RESEARCH

Experts' understanding of writing processes and the best methods for teaching them is disparate and continues to shift. To teach effectively, educators must consequently learn about leading approaches, their underlying assumptions, and best practices for maximizing student learning. The spirit and design of this book reflect our view that developing effective instructional skills for the composition classroom of the 21st century requires, among other things, an explicit awareness of the precepts that guide our beliefs and practices. Knowledge of formal models, theories, and methods enables teachers to discover, build, and articulate their own theories. Explicit knowledge also helps teachers appreciate the strengths and weaknesses of their own skills, investigate and reflect critically on their practices, and become agents of change ([Borg, 2009](#); [Hedgcock, 2002, 2009](#); [Larsen, 2013](#); [Snow, 2005](#); [Tardy, 2017](#); [Tsui, 2009](#)).

In tandem with the insights of empirical research, formal theories therefore can – and should – play a vital role in our instructional planning, teaching practice, and assessment processes ([Seloni & Lee, 2019](#)). To make the most of theoretical models and formal research, however, we must appreciate the complex relationship between instructional practice and theoretical knowledge, which is so often simplistically characterized in terms of a misleading “theory versus practice” dichotomy. At worst, such false views can generate unnecessary and unjustified mistrust (if not outright antagonism) between “theorists” and “researchers,” on one hand, and “practitioners,” on the other. Observers have critiqued this unfortunate and unproductive dualistic viewpoint, which can lead pre- and in-service teachers to see instructional practice to be somehow atheoretical and far removed from formal theory ([Atkinson, 2010](#); [Cumming, 2010](#); [Hedgcock, 2010](#); [Zhu, 2010](#)).

At best, this dichotomous perspective promotes the misguided assumption that classroom teachers must “apply” theoretical models, which should rigidly drive instructional processes.

Of course, teachers do not simply “apply” a theory or method and then magically ensure that students learn something. Nevertheless, we firmly believe that theoretical knowledge is central to how we teach writing. As teachers, we benefit from understanding the enormous practical utility of theoretical models and empirical research available to us. Without the knowledge provided by theoretical principles, we lose sight of crucial tools for responsible instructional planning, effective classroom decision-making, productive expert response, and meaningful assessment. It is worth recalling an oft-cited dictum attributed to the pioneering social psychologist Kurt Lewin: “There is nothing so practical as good theory” ([Lewin, 1951](#), p. 7). Arriving at a single, unified, and “good” theory is probably illusory – and not necessarily desirable. At the same time, the range of theoretical and research-based sources available in L2 writing provides questions and insights of profound value to teachers, as we discuss in [Chapter 3](#). Empirical questions of particular interest to L2 writing educators include the following:

1. What is writing?
2. How do writers generate texts, and why do readers consider certain texts to be effective?
3. How and why does writing quality vary when context, topic, and task are held constant?
4. Why do some writers produce successful texts under certain conditions and when assigned specific composing tasks?
5. How and why do writers’ developmental paths vary under different conditions?
6. What frameworks should we use to construct effective, construct-appropriate writing curricula?
7. What are the best criteria for evaluating curricula, teaching methods, and educational outcomes?
8. How do we devise productive teaching practices and response methods?
9. How do we devise reliable, valid, and fair procedures for assessing written products and writers’ skills?
10. In what ways do culture, socioeconomic structures, educational systems, and writer identity influence teaching and learning?

(Sources: [Belcher, 2012](#); [Hedgcock, 2010](#); [Hyland, 2019](#); [Riazi et al., 2018](#)).

In this chapter, we first sketch a background for subsequent chapters, which explore these areas of concern and survey theoretically informed, research-based strategies for providing effective L2 composition in diverse educational contexts. After presenting an overview of the origins and nature of writing, we examine features of L2 writing that distinguish it from writing in one’s maternal language(s).

FUNDAMENTALS OF WRITING AND WRITING SYSTEMS

Writing is magical, mysterious, aggressive, dangerous, not to be trifled with. Although it takes many forms, it is always a technology of explosive force, a cultural artifact based not in nature ... but sprung from the human mind.

([Powell, 2012](#), p. 11)

[Powell's \(2012\)](#) characterization of writing evokes a powerful set of meanings. It reminds us that, even though members of literate societies view writing as a commonplace tool for communication, record-keeping, and learning, it should not be taken for granted. Writing is a unique system because – rather than directly representing objects, events, and emotions – it uses recognizable symbols “to represent many types of information (e.g., words, initial sounds of words, concepts, categories of objects, grammatical elements)” ([Seidenberg, 2017a](#), p. 37). This symbolic relationship distinguishes written communication from speech in significant ways. Specifically, oral language use involves universal cognitive skills, whereas writing systems and practices are human inventions that must always be transmitted through social interaction. Children naturally develop the ability to understand and produce speech without instruction. On the other hand, “*we were never born to read,*” as the human brain and visual system did not actually evolve to process written language; learning to write and read therefore requires explicit teaching ([Wolf, 2016](#), p. 3, italics in original). Writing represents a profoundly influential technological innovation that serves modern cultures in easily overlooked ways: Writing allows users to control messages and the flow of information while it simultaneously shapes knowledge, how we process it, how we store it, and how we retrieve it ([Birch & Fulop, 2021](#); [Dehaene, 2009](#); [Squires, 2020](#); [Willingham, 2017](#); [Wolf, 2016, 2018](#)).

Before delving into writing processes and the theories that explain them, we should first establish an operational definition of writing as a system that combines semiotic, communicative, cognitive, and creative functions. [Coulmas \(1996\)](#) proposed that a writing system consists of:

a set of visible or tactile signs used to represent units of language in a systematic way, with the purpose of recording messages which can be retrieved by everyone who knows the language in question and the rules by virtue of which its units are encoded in the writing system.

(p. 560)

[Powell \(2012\)](#) similarly defined writing as “*a system of markings with a conventional reference that communicates information*, like the signs on this page” (p. 13, italics in original). [Seidenberg \(2017a\)](#), p. 46) succinctly captured the underlying logic reflected in these definitions:

1. Spoken languages use sound (phonology) to represent meaning (semantics);
2. Writing systems represent spoken languages; therefore,
3. Writing systems represent sound (phonology) and meaning (semantics).

With this general understanding of the relationships between oral and written language in mind, we should specify three levels of description used to characterize and classify writing systems:

- A *writing system* consists of units of oral language (sounds, syllables, words) represented by physically perceptible (visual or tactile) symbols called *graphs* or *graphemes*. For example, the Roman, Greek, and Cyrillic alphabets consist of graphemes, or letters, that correspond to individual speech sounds or sound clusters. Such alphabets are called *phonographic* or *glottographic* because their graphemes symbolize phonemes and how people produce them phonetically. Phonographic systems are sometimes described as the most *granular* because they represent the smallest unit of oral language, the phoneme. In contrast, the inventories of *syllabic* writing systems like Japanese *kana* and Korean *hangeul* mainly (though not exclusively) include graphemes that correspond to syllabic units, which are comparably less granular than phonemes. *Logographic* and *morphographic* (meaning-based) writing systems such as Chinese *hanzza* are even less granular, as they largely feature semantic (meaning-bearing) constituents such as morphemes (word parts), whole words, and word clusters, while also incorporating phonographic representations. We should keep in mind that these categories are far from absolute, as some writing systems incorporate features of more than a single classification, as is the case with Chinese (which is not purely logographic, contrary to popular belief).
- The particular way in which users of a given language implement a writing system is known as its *orthography*, or spelling. Users of European languages such as Croatian, Danish, Dutch, English, French, German, Italian, Polish, and Spanish (to name a few) and of non-European languages such as Turkish use the Roman alphabet to write. The Cyrillic alphabet allows users of Bulgarian, Mongolian, Russian, Serbian, and Ukrainian to read and compose texts in those languages. A language's orthography enables its users to produce written messages based on its particular rules for mapping written markings onto speech and vice versa. From this small sample of languages, we can see that the same alphabet or graph-eme set can be used to write in multiple languages. Writers of English and Spanish use the very same alphabet, but they must learn spelling rules unique to each language's orthography. Orthographic systems vary with respect to the degree of correspondence between symbol and speech, a dimension known as *transparency* or *depth*. Writing systems in which graphemes reliably predict their corresponding sounds are said to have *transparent* orthographies, as in the case with Korean, Spanish, and Thai. In contrast, *opaque* orthographies involve symbol-to-speech mapping rules that are comparatively less predictable and thus more complex, as is the case with Arabic, English, French, and Hebrew.
- Though definitions can vary, *script* refers to the font and distinct visual features of a writing system's markings (letters, characters, symbols) and printing conventions. *Hanzza* script (*sinograms*) are used to write Mandarin, Cantonese, and other regional varieties of Chinese. Both *hanzza* and *hangeul* represent Korean.

Devanāgarī script is used to write Hindi. Scripts can reflect distinct styles (e.g., block, cursive, and so on) and can be written in different directions (e.g., left-to-right, right-to-left, vertically, and so forth). The text that you are now reading, left-to-right, is a Roman-based script (the Roman or Latin alphabet) in a font known as Bembo. (Adapted from [Hedgcock & Ferris, 2018](#), pp. 15–17).

These important terminological distinctions should demonstrate that writing systems can vary considerably in terms of their particular character inventories and visual representations, the units of language symbolized by their markings, and the conventions for producing meaningful written messages. We believe that it is not just helpful but crucial for writing professionals to recognize the diverse ways in which writing can represent speech and convey meaning. It is also essential for teachers of writing to appreciate the ingenuity of writing as a complement to – and extension of – spoken language. By examining the diversity of writing systems and their evolution over the past five (or so) millennia, we can also understand how written communication has shaped social interaction and thinking itself ([Baron, 2009](#); [Lock & Gers, 2012](#); [Olson, 2012](#); [Wolf, 2007, 2016, 2018](#)).

Origins of Writing

Writing is the most important technology in the history of the human species, except how to make a fire.

([Powell, 2012](#), p. 11)

[T]he creation of writing was one of the greatest achievements in human history ... Without writing, there would be no printing press, lightbulb, computer, or Internet.

([Seidenberg, 2017a](#), pp. 31–32)

It is common for people to assume that writing is an old or even ageless human tool, but symbol systems that we frequently call *writing* systems or scripts involve writing connected to speech (*lexigraphic writing*). Writing systems and scripts are quite recent – dating to about 3500 BCE. Writing scholars (anthropologists, archaeologists, and linguists) draw a sharp distinction between scripts and their antecedents, which consist of representational images (*pictograms*) of human figures, animals, plants, the sun, stars, hunting expeditions, and so on. Such pictorial systems, which humans have been creating for over 30,000 years, were precursors of writing, called *protowriting* ([Coulmas, 1996](#); [Dehaene, 2009](#); [Powell, 2012](#); [Rogers, 2004](#); [Seidenberg, 2017a](#); [Wolf, 2007](#)). In a pictorial system, an outline of a bird or fish could represent their living counterparts, as we see in the Sumerian pictograms shown in [Figure 1.1](#). Exact representation in a pictorial system is not required, as long as images are clear enough to be recognized as part of a narrative. To “read” a sequence of pictograms, a person would only need to know the symbols; the order of pictograms could be described orally in any language. This kind of system is known as *primary symbolization* or *semasiography*, protowriting in which signs are not linked to speech forms and can appear in any sequence. As we can imagine, the ambiguity of a string of

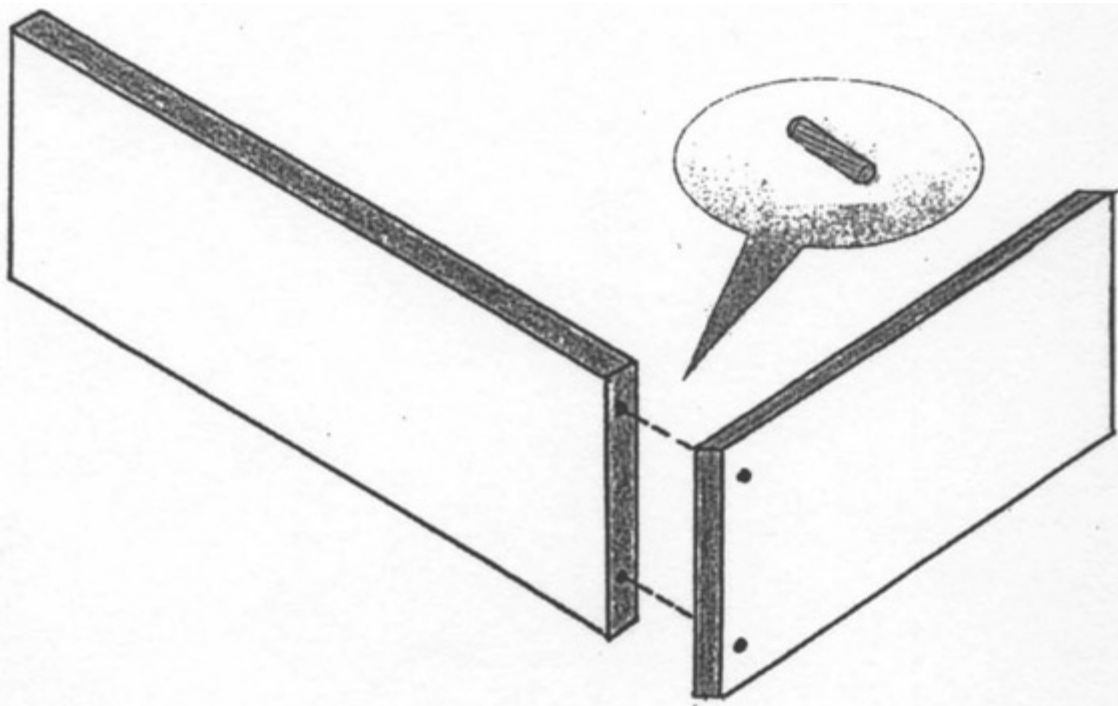
pictograms could make interpreting the intended message rather tricky, as the symbols would not have to be directly linked to words, morphemes, syllables, or phonemes in any given language ([Powell, 2012](#); [Rogers, 2004](#); [Seidenberg, 2017a](#); [Sproat, 2020](#)).

	bird
	fish
	sun
	grain
	orchard

[Figure 1.1](#) Sample Sumerian pictograms.

Archaeological evidence suggests that pictorial systems may have emerged as early as the Paleolithic Era (extending from roughly 2.6 million years ago to the end of the Pleistocene Age, about 10,000 BCE). *Petroglyphs*, graphic symbols etched on boulders with small stones, have been dated to around 100,000 years ago, as have mnemonic (memory) tools for storing information. Often described as a form of primitive art, the celebrated cave paintings of Lascaux in France’s Dordogne region dating to about 20,000 BCE are also forerunners of writing. As with comparable but more recent petroglyphs discovered in the Utah desert and at other archaeological sites throughout the world, “we can never understand these remnants of the past because we do not know why they were made or what they meant to their maker” ([Powell, 2012](#), p. 21).

A second category of protowriting is the *descriptive-representational* device or picture, which tells the viewer something by relying on some convention of figurative art, as in the illustrated instructions for assembling a shelving unit in [Figure 1.2](#). Readers will note that the image *shows* but does not use a script or linguistic signs to *tell* the user how to put the components together, a kind of iconic representation that [Olson \(2012\)](#) called *tokening reality*. Similarly, descriptive-representative artifacts such as the Narmer Palette, discovered at Hierakonpolis in southern Egypt, demonstrate the ancient Egyptians’ effort to communicate complex material and tell stories by systematically arranging conventionalized figures ([Powell, 2012](#)). Dating from 3100 BCE, the beginning of the Pharaonic period, the Narmer Palette consists of an array of figures depicting a battle between north and south Egypt, as well as marks believed to index human speech forms (*hieroglyphs*). The Narmer Palette and similar archaeological artifacts thus present “different ways of imparting information ... happily rolled into a single document” ([Powell, 2012](#), p. 29).



[Figure 1.2](#) Descriptive-representational illustration.

A third category of protowriting, the *identifying-mnemonic* device, is a precursor to writing that clearly establishes a protagonist or event by isolating a particularly obvious or noteworthy feature of a person or event ([Powell, 2012](#)). Identifying-mnemonic devices, which were common among ancient cultures across the globe, might consist of drawings of human figures to represent individuals, animal figures in a narrative, or symbolic depictions of events. For example, the Winnebago Indians devised a symbol system to keep track of songs, which recounted events such as seasons, migrations, battles, and the like. Preserving records by painting images on muslin cloth, Northern Plains Indians such as the Lakota kept track of time and recorded historical events by maintaining a count of winter seasons. Under the direction of tribal elders, a tribal record-keeper might assign a year a specific image embodying a signature event associated with that season, such as a drawing of a teepee symbolizing a migration or a figure of a deer representing a successful hunting expedition. These “picture-writings” consisted of winter counts arranged in sequences running from the lower-left corner of a cloth to the right, then turning the corner and running right to left. By referring to these linear sequences of mnemonic symbols, record-keepers could retell events in oral narratives and songs. However, unlike the lexigraphic scripts familiar to us today (alphabets, syllabaries, and logographic systems), picture writings could not establish an absolute point in time or space. Moreover, records of winter counts could not express equally intelligible messages across tribal boundaries, as each record-keeper devised his own set of local mnemonic conventions. Picture writings, as [Powell \(2012\)](#) observed, were “private symbol systems” that could not “refer to elements of speech” (p. 31).

These protowriting systems are numerous and were widely distributed geographically, in locations as remote from one another as China, Egypt, and Mexico. For the most part, protowriting systems emerged independently. In many instances, they were invented as hunter-gatherer societies became agrarian and more permanent. One of the most well-documented cases of such an innovation took place in Mesopotamia (near the Tigris and Euphrates rivers, in modern-day Iraq and Iran) as early as 8000 BCE, where the Sumerians inscribed wedge-shaped strokes (*cuneiform*) and stylized lines on clay tablets, which were counting tokens used to record financial transactions. Users carried counting tokens in clay envelopes called *bullæ*, which the Sumerians

eventually inscribed with the type and number of tokens inside – an ancient predecessor to the electronic spreadsheets used by modern accountants.

These *bullae* inscriptions were thus symbolic representations of other symbols, making this a *secondary symbolization* system that led to the next major step in the progression toward a full-fledged writing system. For example, consider the Sumerian symbol (*glyph*) for arrow:



Realized as [ti] in speech, this glyph represented the word for life, *til*.² With increasing and more widespread use, glyphs would gradually become associated with the sounds of the words that they represented, leading to the creation of more and more symbols representing a growing number of referents ([Ager, 2021](#); [Schmandt-Besserat, 2012](#)). [Seidenberg \(2017a\)](#) proposed this simple diagram to illustrate this pivotal shift from signifying objects to representing language:

Token → Label → Word

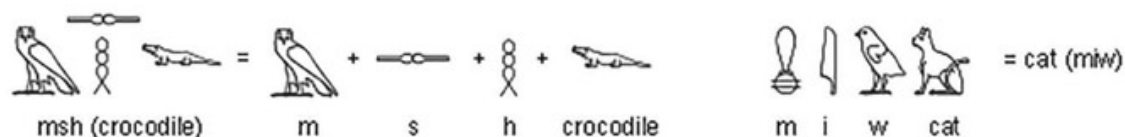
Seidenberg called this step forward “the great enabling advance in the development of writing systems” and “a great leap forward, initiating several thousand years of activity that resulted in modern writing systems” (p. 41). As the token system proliferated, the form of the token (an object) became disconnected from the form of its referent (another object). “Thus language provided one of the key building blocks of the token system, which eventually led to writing, a system for representing language. Perfect!” ([Seidenberg, 2017a](#), p. 44).

As the number of symbol-referent relationships multiplied, scribes discovered the *rebus principle*, a crucial development on the way to lexigraphic writing ([Powell, 2012](#); [Wolf, 2007](#)). A *rebus* is a shortcut that establishes a direct correspondence between symbol and speech, increasing the efficiency of a script. For instance, the rebus principle applies when the Sumerian symbol for “sun” (see [Figure 1.1](#)) signifies not its referential meaning but rather its sound, the word’s first syllable. Because many glyphs could be pronounced the same way while representing different words, a system of *determinative* forms emerged. Determinatives are markings that index a word’s semantic category and phonetic constituents; they show how to pronounce it, distinguish it from its *homophones* (words with the same pronunciation), and thereby minimize ambiguity.

Leading writing scholars maintain that when a writing system begins to apply the secondary symbolization of the rebus principle to all sounds, it reaches a sort of critical mass or tipping point ([Crystal, 2006](#); [Seidenberg, 2017a, 2017b](#); [Wolf, 2007](#)). The systematic use of phonograms combined with the construction of sentences based on the syntax of oral language triggered “the true takeoff of writing” ([Schmandt-Besserat, 2012](#), p. 9). After the application of secondary symbolization, a script can efficiently and systematically encode a great number of meanings. In the case of Sumerian cuneiform, which is primarily syllabic, scribes soon realized that they could combine graphemes (representing single syllables such as [ti]) to form compounds. Through compounding, scribes could express even more complex meanings and relationships, ultimately constructing phrases and clauses. These discoveries made a momentous historical impact. As [Powell \(2012\)](#) noted, writing “cast a veil across the human past, separating the million human years that came before from the turbulent last five thousand years.” The Sumerians discovered the phonetic principle in graphic representation around 3400 BCE – a relatively short time ago. Since that epiphany, “when conventional markings first represented sounds of the human voice – the

cultures encoded in this and subsequent related traditions of writing have changed human life forever” (p. 11).

Egyptian hieroglyphs similarly took form as pictograms (glyphs) (cf. [Figure 1.1](#)) and gradually developed into a complex script that comprised *logograms* (which represent morphemes), *phonograms* (which represent consonants and consonant clusters), and determinatives (semantic classifiers). Many Egyptian logograms reflected the pictograms from which they evolved: From these symbols, scribes constructed the world’s first alphabet by the *acrophonic principle*, which allowed for a hieroglyph for one word to represent the first consonant of that word – with no connection to the word’s meaning. For instance, as we see in [Figure 1.3](#), the glyph for crocodile consists of a figure of a crocodile, which represents the phoneme cluster /mʃ/. Like the combinatorial principle that emerged in Sumerian, Ancient Egyptian combined the glyph for crocodile with the glyphs that spell out /mʃ/. Likewise, the hieroglyphs for cat merge the glyphs for /m/, /i/, and /w/ with an image depicting a cat ([Ager, 2021](#)). The establishment of a direct connection between visual symbols and sound symbols was a natural precursor to alphabetic systems, which we recognize as a familiar category of *writing* today ([DeFrancis, 1989](#); [Diringer, 1968](#); [Sproat, 2020](#)).



[Figure 1.3](#) Egyptian glyphs for *crocodile* and *cat*.

(Adapted from: www.omniglot.com/writing/egyptian.htm)

Lexigraphic Writing

Reading and writing are among the towering achievements of human cognition and are the hallmarks of modern civilization.

([Yap & Liow, 2020](#), p. 453)

Just as the dating of protowriting and writing systems is hardly an exact science, classifying forerunners of writing and the lexigraphic scripts that emerged from them can be equally challenging. Writing scholars generally agree that lexigraphic writing systems probably took shape around 3500 or 3400 BCE, evolving independently of one another at distinct times and in different locations around the world, including Mesopotamia, China, and Meso-America, among others. As we will see later in this chapter, these systems share common features. Nonetheless, contrary to popular belief, archaeological evidence in no way supports a “common origin” theory ([Crystal, 2006](#)). Contact among civilizations precipitated by historical events (i.e., exploration, trade, conflict, conquest, colonization, migration, and so on) often led to language change and consequent cross-cultural influences on writing and protowriting systems (where they existed).³ Contact between two or more cultures can necessitate the emergence of a *lingua franca*, as well as pidgins and creoles ([Holmes & Wilson, 2017](#)). Sustained contact can similarly contribute to changes in literate practices and writing systems. For example, before declining into disuse, Egyptian hieroglyphic script had already paved the way for a new writing system, the progenitor of the Roman alphabet. As Egyptian influence receded, the Phoenician, Greek, and Roman

cultures rose in stature and propelled the shift from hieroglyphics to the Roman alphabet ([Lock & Gers, 2012](#); [Powell, 2012](#); [Seidenberg, 2017a](#)). Given the dynamic ebb and flow of intercultural contact around the Mediterranean, it should be easy to see that the boundaries between and among scripts can sometimes blur, like the classification systems developed by writing scholars.

A detailed study of the world's writing systems is not possible here, but it is useful for writing teachers to acquaint themselves with the formal features that distinguish them from one another. (For thorough and interesting accounts of the history of writing, consult Further Reading and Resources at the end of this chapter, as well as the Omniglot web-site: www.omniglot.com.) For example, L2 writers who already know a writing system that differs considerably from that of the target language may require special assistance in recognizing elements that may or may not transfer successfully across scripts and orthographies ([Cook & Bassetti, 2005](#); [Hedgcock, 2012](#); [Koda, 2005](#); Koda & Zehler, 2007).

A fundamental property shared by lexigraphic writing systems is that their symbols correspond to speech and reflect the natural sound sequences of oral language. As [Willingham \(2017\)](#) observed, “[i]nstead of writing down thoughts, we write down oral language. Writing codes sound” (p. 23). Lexigraphic scripts contain symbols (*graphs*) that signify linguistic units, such as words, morphemes, syllables, phonemes, phonemic features (e.g., voicing, tone), and phonetic features (e.g., aspiration and syllable stress, which may be predictable by phonological rule). In some writing systems, graphs can represent combinations of these units (e.g., a word or syllable plus a tone). Words and morphemes express meaning, whereas syllables, phonemes, and features do not convey meaning on their own. For example, like *phonemes* (which represent distinct sounds), *graphemes* “are the smallest units in a writing system capable of causing a contrast in meaning” ([Crystal, 2006](#), p. 105). The alternations that we can see and hear among the English words in the minimal set *bad*, *dad*, *fad*, *had*, *lad*, *mad*, *pad*, *sad*, and *tad* introduce salient meaning distinctions. Thus, , <d>, <f>, <h>, <l>, <m>, <p>, <s>, and <t> signify distinct graphemes ([Crystal, 2006](#); [Dehaene, 2009](#)). These correspondences reflect the *phonographic matching principle*, which holds that “there should be a viable mapping relating spellings to sounds and back,” although uniform one-to-one mapping is rare except in highly transparent orthographies ([Ryan, 2020](#), p. 54).

As we consider how the elements and architecture of writing systems influence writing, reading, and cognition, it is helpful to recognize how scripts contrast, blend, and complement one another (see [Figure 1.4](#)). Chief among the differences is the set of linguistic units represented by a script's graphs. We can assign scripts to two broad but not absolute categories, according to the relationship between their symbols and the linguistic units that they represent: (1) *logographic* systems (*logographies*) contain inventories of graphs that represent morphemes, words, and concepts; and (2) *phonographic* systems (or *phonographies*) consist of inventories containing graphs representing units of sound. We can divide the phonographic category into two overlapping subcategories: *Syllabic* orthographies (*syllabaries*) consist of graphs representing syllables, whereas *alphabetic* scripts (*alphabets*) consist of graphs representing individual phonemes (i.e., consonants and vowels) ([Sproat, 2020](#); [Willingham, 2017](#); [Wolf, 2007, 2016](#)). The non-absolute, overlapping categories listed in [Figure 1.4](#) represent an admittedly imprecise continuum, moving from meaning-based (logographic and lexigraphic) to sound-based (syllabic and alphabetic, or *full*) writing systems, which may intersect. Writing systems and orthographies may be very difficult to classify, as they may exhibit both logographic and phonographic characteristics that function harmoniously together. For instance, contemporary Chinese writing

combines distinct *logographs* (characters) with one another through compounding; these compound logographs are complemented by symbols for tones that signify distinct vowels ([Seidenberg, 2017a](#); [Tsu, 2022](#)).

Category (Relative Transparency)	Language	Script Name	Text Direction	Sample Transcription
Logographic + Logosyllabic (Opaque)	Chinese	Sinograms (<i>Hanzza</i> , or <i>Hanzi</i>)	Left to right, horizontal; right to left, horizontal; vertical, top to bottom	我的老師寫了一本流行的書。
Logographic + Syllabic (Opaque and Transparent)	Japanese	<i>Kanji</i> , <i>Katakana</i> , <i>Hiragana</i>	Right to left, horizontal; vertical, top to bottom	私の先生は人気の本を書いた。
Alphasyllabic (Transparent)	Hindi	<i>Devanāgarī</i>	Left to right, horizontal	मेरे शिक्षक ने एक लोकप्रिय पुस्तक लिखी।
Alphasyllabic (Transparent)	Korean	<i>Hangeul</i>	Left to right, horizontal (since 1980s); right to left, horizontal; vertical, top to bottom	선생님은 인기있는 책을 썼습니다.
Alphasyllabic (Transparent)	Thai	Thai alphabet	Left to right, horizontal	ครูของฉันเขียนหนังสือยอดนิยม
Consonantal-Alphabetic (Opaque)	Arabic	Arabic alphabet	Right to left, horizontal (numerals written left to right)	كتب أستاذي كتاباً مشهوراً.
Consonantal-Alphabetic (Opaque)	Hebrew	Hebrew	Right to left, horizontal	המורה שלי כתב ספר פופולרי.
Alphabetic (Transparent)	Russian	Cyrillic alphabet	Left to right, horizontal	Мой учитель написал популярную книгу.
Alphabetic (Transparent)	Greek	Greek alphabet	Left to right, horizontal	Ο δάσκαλός μου έγραψε ένα δημοφιλές βιβλίο.
Alphabetic (Transparent)	Spanish	Roman alphabet	Left to right, horizontal	<i>Mi profesor escribió un libro popular.</i>
Alphabetic (Opaque)	French	Roman alphabet	Left to right, horizontal	<i>Mon professeur a écrit un livre populaire.</i>
Alphabetic (Opaque)	English	Roman alphabet	Left to right, horizontal	<i>My teacher wrote a popular book.</i>

Figure 1.4 Comparison of selected lexigraphic writing systems and scripts.

(Adapted from: [Birch & Fulop, 2021](#); [Hedgcock & Ferris, 2018](#); [omniglot.com](#))

We should further note that, because scripts are essentially manufactured and therefore adaptable, users of a language may borrow graphs from scripts other than the one originally used to write messages in that language. For example, we all use nonphonemic graphs when we write Arabic numerals, mathematical symbols, and punctuation marks such as 1, 2, 3, +, −, =, <, >, %, ¢, \$, £, €, ¶, ?, !, and so on. When we use a keyboard, these marks are available as ASCII and Unicode characters. Sometimes called *sematograms*, these graphs are not attached to forms of speech, although the symbols themselves have names: We may call @ “the *at* sign” or “the commercial *at*”; & is called an *ampersand*, and so forth. Readers and writers of any language can readily use these logograms without needing to know their names. The combination of sematograms and the lexigraphic symbols that comprise a writing system’s script (letters, logographs, and so on) is sometimes called a *graphabet*, the entire character set available to a writer at a given moment ([Shortis, 2020](#)).

Logographic Writing

In contrast to systems that incorporate logograms to be used alongside their primary scripts, most logographic orthographies align with the structural properties of the spoken language that their graphological inventories represent. For instance, Chinese script, the basis for Japanese *kanji* and Korean *hanzza*, consists largely of logograms (*characters* or *sinograms*, known in Chinese as *hanzi*) that signify morphemes and words.⁴ Logographic writing efficiently represents spoken Chinese dialects and their isolating morphological systems, in which each word is likely to be a single, isolated morpheme. For example, the Chinese equivalent of the English noun phrase *the cat’s tail* would be something like *the tail of the cat*. In *the cat’s tail*, the possessive <-’s> is

directly affixed to the headword <cat>, an example of inflectional morphology. In contrast, *the tail of the cat* is an analytical structure where the morphemes expressing the possessive relationship are isolated from one another. Chinese thus lacks extensive inflectional morphology, a feature that partly accounts for why the Chinese sinogram inventory is thought to number about 60,000 unique forms. At the same time, approximately 80% of Chinese sinograms consist of a *radical* or *signific* – one of perhaps 200 root graphs representing words or bound morpheme – plus a phonetic complement or determinative that tells readers how to pronounce the word.

We can appreciate how Chinese writing relies on both semasiography (a direct connection between sign and meaning) and *phonography* (“sound-writing,” where signs index sound) by considering an example. If we carefully scrutinize the *hanzi* character 案, which refers to *table* or *desk*, we can see the radical 木 embedded in it. The character 案 functions as a single grapheme, pronounced [àn] (with a falling tone) in Beijing Mandarin. The radical 木 can function as an independent graph, a logogram pronounced as [mù] in Beijing Mandarin. When 木 occurs as the radical in 案, however, the radical has no phonetic value; it means “something made of wood” but does not indicate how to pronounce the word. The other constituent of 案 is the Mandarin graph 安, which is pronounced [ān] (with a level tone) and which means *peace* when it stands alone. The sinogram 案 (pronounced [àn], with a falling tone) therefore derives from two graphemes, one symbolizing sound and the other representing its semantic class (“things made of wood”). Because sinograms such as 案 often express approximate meanings and pronunciations, readers of Chinese *hanzi* must memorize (or guess) the sound of the phonetic complement and associate the graph with an already familiar word in order to discern a meaning. Variable pronunciations associated with phonetic complements further complicate the guessing game, which requires Chinese readers to connect spoken syllables and words with sinograms with particular meanings (Birch & Fulop, 2021; Powell, 2012; Seidenberg, 2017a; Tsu, 2022). Form-meaning correspondence in Chinese script is nonetheless predictable enough to allow speakers of diverse, mutually unintelligible regional dialects (e.g., Cantonese, Hakka, Hokkien, Mandarin, Min, Xiang, and so on) to communicate easily using the same set of characters, whose forms may have changed little since their invention.

Although estimates vary, research suggests that a reader of Chinese must know as many as 6,600 sinograms in order to comprehend most texts, but that the 1,000 most frequent sinograms make up about 90% of texts aimed at a general audience. A skilled reader of scholarly and literary texts may require a mental inventory as large as 30,000 sinograms (Mair, 1996). To put estimates of required vocabulary size into perspective, comprehensive Chinese dictionaries may include around 50,000 *hanzi* (entries), most of which represent low-frequency words. These demands on memory are formidable, but it is instructive to recall that dictionaries of modern English may contain up to a half-million separate entries. The *Oxford English Dictionary* tops out at about 600,000 words. At the same time, recent neurolinguistic research demonstrates that becoming literate in Chinese places unique demands on regions of the brain that house lexigraphic memory, influencing neuronal circuitry (Dehaene, 2009; Seidenberg, 2017a; Tsu, 2022; Willingham, 2017; Wolf, 2007, 2016, 2018). Knowledge of a particular script can affect learning processes – including the learning of a new writing system – in complex ways. Consider, for example, a literate Chinese speaker learning English, which is written in the Roman alphabet: Some dimensions of reading and writing in English may seem easy, whereas developing necessary strategies such as word analysis and decoding might be quite challenging (Hedgcock, 2012; Hedgcock & Ferris, 2018; Leong, 1995; Nation, 2013; Newton et al., 2018; Venezky, 1995).

Syllabic Writing

Unlike logographic scripts, which rely largely on form-meaning correspondences, syllabaries mainly comprise graphemes that represent syllables. *Alphabetic-syllabaries*, as their name suggests, are hybrid systems whose scripts consist of graphemes representing both syllables and phonemes. The Korean *hangeul* grapheme (called *yae*), for instance, represents two adjacent phonemes that comprise the syllable [jæ] in spoken Korean. Syllabic scripts such as *hangeul*, Mesopotamian, and Egyptian evolved from logographies, with some of their graphemes still resembling historical logograms. Other syllabaries, such as Japanese *kana* and the Cherokee and Tamil syllabaries, associate single graphemes with consonant-vowel (CV) and consonant-vowel-consonant (CVC) clusters, which have natural beats and rhythms in speech and which are indivisible – very much like the letters of an alphabet ([Birch & Fulop, 2021](#)). The Japanese *kana* system, which contains 47 graphemes, consists of two subsystems, *hiragana* (which represents grammatical functions) and *katakana* (symbols used to transcribe non-Japanese forms). *Kana* graphemes, which largely represent grammatical inflections and particles, allow users to transcribe the 100 or so syllables that occur in spoken Japanese. Meanwhile, Chinese-derived *kanji* logograms require users to develop a lexical memory that directly associates individual characters with meanings. Literacy in Japanese thus requires mastery of three interrelated scripts, which are intermixed in ordinary texts. Learning Japanese writing requires a combination of memory, recognition, decoding, word analysis, and interpretation competencies ([Mori, 1998](#); [Seidenberg, 2017a, 2017b](#)).

Alphabetic Writing

Korean *hangeul* and Thai scripts similarly draw on two types of orthography, as their grapheme inventories include both syllabic and phonemic symbols. These scripts are sometimes called *alphasyllabaries* or *abugidas*. “Pure” alphabets rely solely on the *alphabetic principle* (sometimes called the *phonemic principle*), a discovery often attributed to the Ancient Greeks and Romans ([Janson, 2012](#)).⁵ The alphabetic principle holds that each graph (letter) can signify a single sound or syllable and that grapheme sequences form words ([Birch & Fulop, 2021](#); [Dehaene, 2009](#); [Seidenberg, 2017a](#); [Wolf, 2007](#)). Experts such as [DeFrancis \(1989\)](#) have claimed that *full* writing systems must be essentially phonological, “with the main distinction being between syllabic and segmental systems, and within the latter, consonantal and alphabetic” ([Sproat, 2020](#), p. 29).

Of course, no alphabetic script adheres strictly to a one-to-one relationship between grapheme and phoneme. As a single grapheme can represent multiple sounds, a small alphabet can represent a surprisingly large number of sounds and sound clusters, especially with the addition of *diacritical marks* (e.g., accents [ˈ], umlauts [¨], tildes [~], and so forth). Most alphabets were designed and tailored to capture the phonemes and phonological rules of particular languages or language groups. Thanks to the phonographic principle, many alphabets can be used to write a range of languages. However, for a writing system to be effective, its properties “must align with the properties of the spoken language it represents” because “a unit that is well suited to one type of language may be disastrously bad for another” ([Seidenberg, 2017a](#), p. 52). Writers of English, French, Spanish, German, and many other predominantly consonant-vowel (CV) languages use the Roman (Latin) alphabet because the alphabetic symbols harmonize with the small size of

speech units. Speakers and writers of Russian, Ukrainian, Bulgarian, and other Slavic languages use the Cyrillic alphabet for similar reasons. Arabic and Hebrew scripts represent consonantal alphabets (*abjads*), which require readers to reinsert (*interdigitate*) the “missing” vowels ([Bauer, 1996](#); [Birch & Fulop, 2021](#)). Readers of abjads read the consonant template (a predictable sequence of consonants such as <k-t-b>), connect the template with a semantic set (in the case of <k-t-b>, the general topic area of literacy, reading, writing), and examine the surrounding text for further clues about local meaning and the general topic. The consonantal alphabets of Arabic and Hebrew provide all the clues needed for writers and readers to encode and decode meaning; vowels are unnecessary and would provide too much information ([Seidenberg, 2017a](#)). To use alphabetic scripts – whether fully alphabetic or consonantal – readers and writers must learn to relate speech to written words, and vice versa. Obversely, a person could potentially become literate in Chinese without learning to speak the language or understand spoken messages, as Chinese logograms directly represent meaning-bearing constituents (morphemes, words, and even abstract concepts) (see [Figure 1.4](#)).

The degree of phoneme-grapheme (sound-symbol) correspondence varies widely across the world’s orthographies. Research on L2 learning has consistently demonstrated that the similarities and differences between learners’ L1 and L2 orthographic features can influence the ease and speed with which they acquire L2 reading and writing skills. For example, neuroscientific evidence has revealed that “the [neurological] circuit for an English-alphabet-reading brain has both shared components and some different component parts from non-alphabetic orthographies, and even from some other alphabetic ones” ([Wolf, 2016](#), p. 73). Predictably, learners whose L1 orthographies differ from L2 orthographies must process visual and phonological information differently as they learn to read and write in the L2. Specific challenges for L2 readers and writers may include learning new strategies for analyzing, decoding, and encoding text that is represented logographically, syllabically, or alphabetically. At the same time, L2 learners can successfully transfer L1 text processing strategies that are consistent with L2 orthography. Studies have revealed evidence for positive, negative, and non-transfer of L1 orthographic knowledge to the L2. Globally speaking, most L1 → L2 transfer of reading skills across orthographies tends to be positive.⁶

Transparency and the Orthographic Depth Hypothesis

A useful tool for classifying orthographies and for understanding how knowledge of one type of script might influence the learning of another is their comparative *transparency* or *opacity* – the degree to which orthographies adhere to a one-to-one principle. As illustrated in [Figure 1.4](#), we can classify orthographies as *transparent* (*shallow*), *opaque* (*deep*) or – in many cases – somewhere in between. For instance, because of fairly close grapheme-phoneme correspondences that allow readers and writers to “sound out” words and morphemes, the orthographies of Korean (which uses the *hangeul* alphasyllabary), Serbian (which uses the Cyrillic alphabet), and Finnish and Turkish (which are transcribed in the Roman alphabet) are considered to be *transparent* relative to other orthographic systems. That is, they largely adhere to the *phonographic matching principle*, which requires “a viable mapping relating spellings to sounds and back” ([Ryan, 2020](#), p. 54). Relatively less transparent are the orthographies of Italian and Spanish (written in the Roman alphabet) and Modern Greek (transcribed in the Greek alphabet). Even less transparent (more opaque) are German and Swedish orthographies (both written in the Roman alphabet),

owing to their comparably higher degree of phonological irregularity. Generally speaking, transparent (shallow) orthographies are thought to place lower cognitive demands on learners than do relatively deeper (more opaque) orthographies.

Edging further toward the opaque end of the transparency continuum are French and Danish (also transcribed in the Roman alphabet). It will not surprise readers that English orthography is classified as even more opaque than the orthographies of French and Danish, all of which are *psycholinguistically deep*. Because of extensive phonological irregularity in English, learners must master not only predictable sound-symbol correspondences, but also those that are irregular and idiosyncratic to individual words. For instance, how can <right>, <rite>, <wright>, and <write> be spelled so differently yet correspond to a single pronunciation ([ʁajt] or [rajt] in North American English)? “English spelling,” wrote [Sproat \(2020\)](#) “is quirky and unpredictable. Indeed, it probably has the most complicated letter-to-sound correspondence of any language that uses a segmental writing system” (p. 31). Because of its opacity, English spelling certainly poses distinct challenges to readers and writers ([Yule & Yasuko, 2020](#)).

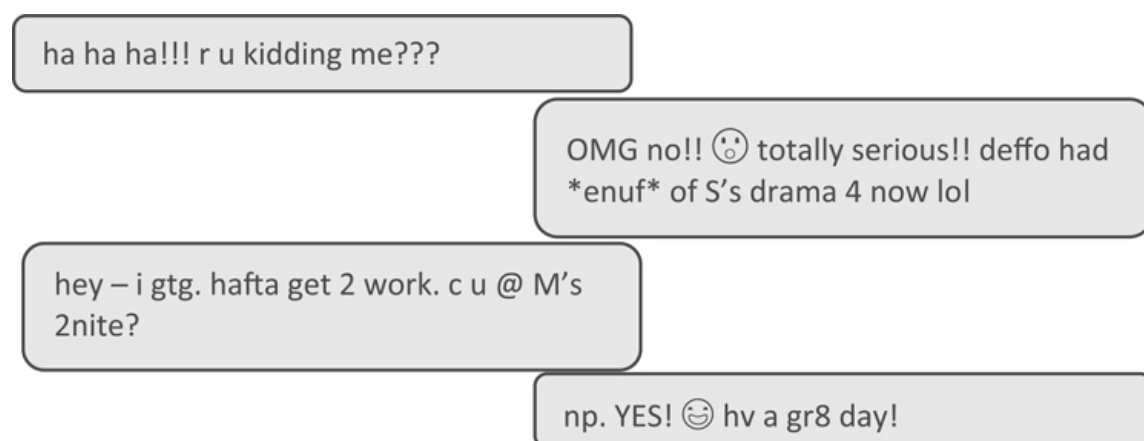
Its opacity notwithstanding, the English orthographic system may not deserve quite so much notoriety for being irregular, arbitrary, and difficult to learn ([Birch & Fulop, 2021](#); [Brown, 2019](#); [Cook, 2004](#); [Hedgcock & Ferris, 2018](#)). We should first recognize that English spelling is primarily *morphological*, rather than phonological or phonemic; it represents “not surface, but underlying phonological representation,” which is more systematic than we might think ([Sproat, 2020](#), p. 33). Moreover, English orthography is partly governed by the *distinctiveness principle*, which holds that “every word should have its own spelling.” English spelling is also influenced by the *invariance principle*, which maintains that much of its unpredictability “arises because spellings never change,” while pronunciations can change a great deal ([Ryan, 2020](#), p. 55). Still more opaque than English orthography are the orthographies of Arabic, Hebrew, and Aramaic, whose consonantal alphabets lead to considerable ambiguity and require readers to insert vowels, as we noted earlier. Among the most opaque (least transparent) orthographies are Chinese *hanzza* and Japanese *kanji*, whose minimal phonographic elements (phonetic determinatives) seldom provide readers with enough information to pronounce words based solely on logograms.

Research on the Orthographic Depth Hypothesis (ODH) ([Katz & Frost, 1992](#)) informs our current understanding of the relative transparency of writing systems. The ODH predicts that shallow (regular, transparent) orthographies such as those of Serbian and Spanish induce learners to analyze words phonologically at the intraword level when they read ([Defior et al., 2002](#)). Conversely, the ODH holds that learners of *deep* (less regular, opaque) orthographies such as French and English must rely more on lexical information specific to individual word forms and less on intraword (morphological) analysis ([Dehaene, 2009](#); [Grabe & Stoller, 2020](#); [Seidenberg, 2017a](#); [Wolf, 2007](#)). Many questions regarding the influence of L1 orthographic knowledge on L2 literacy development remain unanswered, yet this productive line of inquiry helps us understand and appreciate the difficulties faced by students as they develop fluency, accuracy, strategic skills, and a robust vocabulary for L2 reading and writing ([Grabe, 2009](#); [Grabe & Stoller, 2020](#); [Hedgcock & Ferris, 2018](#); [Koda, 2011](#)).

Interactions Among Writing, Speech, Communication, and Cognition: Writing as Technology

Our survey of the origins and evolution of writing aligns with a consensus view that writing fundamentally consists of “*a system of markings with a conventional reference that communicates information*” (Powell, 2012, p. 13, italics in original). To go beyond a clinical description of the contents and structure of writing systems, though, we must consider interactions between writing and speech, as well as the indelible imprint of writing on communication and cognition. If all writing systems were phonographic, we might reasonably conclude that writing amounts to “speech written down” – an optional extension of oral language (which we all acquire without formal instruction). Admittedly, most of the world’s orthographies reflect secondary symbolization and are, to varying degrees, bound to the constituents and rule systems of spoken language. It would nonetheless be misguided to conclude that written communication functions simply as a more permanent *re*-presentation of oral language, or “the graphic counterpart of speech” (Diringer, 1968).

In fact, writing and speech interact synergistically: “[N]o firm line can be drawn between the oral and the written” (Finnegan, 1988, p. 178). Synchronous and asynchronous computer-mediated communication (CMC) is a sphere of written communication where we can readily observe the dynamic interplay of speech and writing. CMC includes modes of mobile communication such as short-text message systems (SMS), online chat, Twitter, and email. A vibrant sphere of linguistic research, CMC has generated many innovative forms, which Crystal (2011) called “internet graphology.” These forms may phonographically represent vernacular speech while flouting orthographic, grammatical, and punctuation conventions, as we see in the brief extract from an SMS exchange shown in Figure 1.5.



[Figure 1.5](#) Extract from sample text-message (SMS) exchange.

This brief extract exhibits several features that are prototypical of digital writing (internet graphology):

- unconventional, missing, or multiple punctuation marks
- unconventional or eccentric spelling
- all capital letters
- emphatic asterisks
- laughter representations
- emoticons or emojis
- abbreviations, initialisms, acronyms, and rebuses

r u kidding me???

enuf, gonna, hafta

OMG, YES

enuf

ha ha ha!!!, lol

😊, 😊

OMG, gtg, 2nite

- lack of traditional capitalization
([Crystal, 2011](#); [Danet, 2001](#); [Squires, 2020](#))

r u kidding me???

Digital writing practices may seem to make written language more directly representative of the oral language produced by speakers ([Squires, 2020](#)). This process reflects speech-to-writing transfer, as we see in (approximate) phonetic spellings such as <enuf> for *enough* and <hafta> for *have to*. Digital typographic practices may also influence spoken language, reflecting writing-to-speech transfer. For example, a person might say “OMG” ([o##ǣm#dʒi]) rather than “Oh my God” or “LOL” ([ε#lo##ɛl]) rather than “Laughing out loud” (which isn’t something that a person would likely say aloud at all). Emoticons, emojis, all-caps, and forms such as emphatic asterisks also indicate that CMC can be “the origination site of written practice that emerges independent of spoken language” altogether ([Squires, 2020](#), p. 481). Although digital writing is derived from oral and traditional written registers, it represents a register all its own.

Designed for speedy use of a keyboard or smartphone keypad, CMC offers adaptable scribal tools for making efficient use of the devices that we rely on for instant communication in the 21st century. We can observe this adaptability in the algorithmically-based auto-correct and auto-type functions of popular messaging apps such as WhatsApp™ and Line™; these predictive functions enable users to type complete messages on a mobile device even more swiftly. CMC has “undoubtedly, and rapidly, changed the social landscape of writing,” as more people than ever before produce prodigious amounts of writing “to be read by many more people than in the past (potentially millions at the touch of a ‘publish’ button)” ([Squires, 2020](#), p. 482). Digital writing has likewise engendered new text forms, transformed genres, and reshaped users’ aims for writing ([Zheng & Warschauer, 2017](#)). The multiple categories of CMC interaction undoubtedly constitute *writing*, although ongoing controversy surrounds the question of whether communication via SMS and Twitter can be classified as a *writing system*. We concur with [Shortis \(2020\)](#), who maintained that “there is written system in texting and other messaging, but it is of a different order” than a *writing system*, which, by definition, is extrapolated from standardized language patterns (p. 487). In other words, CMC’s novel typographic practices and dynamic repertoire of semasiographic and glottographic features reflect the ways in which existing writing systems can adapt to – and flourish in – a new medium. The introduction of movable type printing technology in 11th century China and its more widespread adoption in 15th century Europe similarly supplied a new medium for producing and disseminating written texts. Although print technology undoubtedly influenced writing practices and the ways in which people could learn to read and write, print itself did not constitute a new writing system ([Cook, 2004](#); [Hill, 2020](#); [Olson, 2012](#); [Powell, 2012](#)).

The creation of markings and methods for indexing them to sound and meaning represents a formidable innovation. Scholars often characterize writing itself as a *technology* of immense social and cultural consequence ([Baron, 2009](#); [DeFrancis, 1989](#); [Dehaene, 2009](#); [Powell, 2012](#)). Written messages can be inscribed on stone, clay, wood, cloth, sheepskin, paper, papyrus, or parchment with a stylus, brush, pen, or pencil. Texts can be printed mechanically on paper with a typewriter; they can be mass-produced with a printing press or photocopy machine. We can encode messages digitally as hypertext on a computer keyboard or smartphone and disseminate them on a potentially global scale. Thanks to the ubiquity and efficiency of our devices, apps, and networks, digital writing has become so embedded in 21st century culture that we pay little attention to its origins or complexity ([Barton & Lee, 2013](#)). As [Squires \(2020\)](#) observed:

[T]hat writing is itself a technology can be easy to forget in light of the proliferation of devices and systems now labeled as ‘technologies.’ Writing is certainly a technology, but it is now a quite taken-for-granted one, even as it has been transferred to more media. As print culture has seemingly been replaced with electronic or screen culture, written language is transmitted and viewed largely via screens rather than paper. At the same time, writing has come to be experienced as part of interactive and dynamic interfaces rather than static or monologic ones.

(p. 471)

Regardless of the medium and means of transmission, writing exhibits features that set it apart in significant ways from oral language. Whether recorded on stone, clay, or paper – or encrypted digitally – written texts exhibit features that oral messages clearly lack ([Baron, 2009](#); [Dehaene, 2009](#); [Wolf, 2007](#)). As [Smith \(2004\)](#) emphasized, “spoken language and written language can rarely be the same” (p. 42). Writing has the power to determine the meanings that texts convey, and these meanings may emerge solely within written genres and registers ([Biber, 2009](#); [Shortis, 2020](#); [Squires, 2020](#)). Writing and literacy serve as ingenious tools for communication, meaning-making, and creative expression. Just as importantly, they help us understand human language, the mind, the brain, and society ([Cook, 2004](#); [Crystal, 2006](#); [Olive, 2012](#)). For [Olson \(2012\)](#) and many writing scholars, “writing and reading make unique demands and provide unique opportunities for thinking and, in addition, provide new resources that societies may or may not exploit for various purposes” (p. 39).

WRITING DEVELOPMENT AND THE DISTINCT DIMENSIONS OF LEARNING TO WRITE IN L2 DEVELOPMENT

As our overview of writing systems should demonstrate, developing literacy in any language requires learning not just its orthography and script but also the cognitive functions required to decode and produce written texts. Writing systems must be learned through socially mediated processes, such as formal instruction (usually in school), sustained observation, imitation, practice, socialization, and interaction – an effortful process that typically takes many years. Learning to write in one’s primary language(s) requires early modeling, sustained guidance, and periodic intervention so that neurodevelopmentally ready children can cultivate basic handwriting and spelling skills, as well as reading ability. To nurture a *writing brain*, learners must master visual, psycholinguistic, and memory operations such as symbol recognition and reproduction. These skills underlie the *grapho-motor* faculties (i.e., hand-eye coordination, motor planning, and motor control functions) needed to produce letters, symbols, and logograms manually and with the aid of a keyboard. The ability to construct meaningful print messages relies on automatic control of these functions, along with the robust development of all aspects of oral language skills ([Berninger & Chanquoy, 2012](#); [Richards et al., 2012](#); [Zins & Hooper, 2012](#)).

As literate children advance toward (and through) adolescence and into adulthood, they not only *learn to write*; they also *write to learn*. That is, students learn academic content (e.g., social studies, history, literature, science, mathematics, and so on) partly by writing in those disciplinary

areas. Primary, secondary, and post-secondary education requires student writers to produce high- and low-stakes academic genres (e.g., stories, essays, reports, research papers, and so forth) that presuppose fully automatized graphomotor skills. In addition to building cognitive skills, this trajectory of academic literacy development involves cultivating a writing voice, constructing an identity, and exercising agency with respect to what, how, why, and for whom one writes in a particular language – or in a particular combination of languages ([Bazerman et al., 2017, 2018](#); [Berninger & Chanquoy, 2012](#); [Gee, 2015](#)). Cultivating literate knowledge in an additional language (or L2) can impose relatively greater psycholinguistic, cognitive, and sociocultural demands on learners whose L2 oral and aural proficiency may be emergent and whose literate knowledge in L1 may be limited ([Hedgcock, 2012](#); [Hedgcock & Ferris, 2018](#); [Kanno & Harklau, 2012](#)). The remainder of this chapter reviews dimensions of L2 literacy that distinguish L2 literacy development, L2 writing processes, and L2 writers' texts from their L1 (primary-language) counterparts. Here and in [Chapter 2](#), we will carefully examine the unique profiles of L2 writer populations.

Prior Knowledge: Implications for Teaching L2 Writing

As discussed at length in [Chapter 2](#), a primary feature distinguishing novice L2 writers from their L1 counterparts lies in the prior experience with literacy and the world that they bring to the composition classroom. Differences in background knowledge and strategic proficiency manifest themselves in numerous ways: in L2 writers' responses to texts and content, in their reactions to L2 composition instruction, and in their familiarity with the rhetorical patterns of academic, professional, workplace, and online discourses. [Figure 1.6](#) summarizes a number of these salient distinctions and their implications for L2 literacy education.

Selected Formal Differences Between L1 and L2 Student Writing

- Divergent preferences for arranging textual elements and sequencing information;
- Differing strategies for constructing arguments (e.g., direct development of the writer's position through deductive reasoning, in contrast to inductive reasoning achieved by indirect development of a position, thesis, or opinion);
- Dissimilar methods of integrating evidence and source material into writing (e.g., extent of textual borrowing and attribution, paraphrase, quotation, stylistic imitation, and so on);
- Distinct approaches to meeting reader expectations and schematic knowledge (e.g., assumptions about how much information the reader wants or needs, the degree to which the reader can rely on implicit textual cues to connect points of information, and so on);
- Divergent methods of marking cohesion and textual unity (e.g., use of pronouns, lexical variety and repetition, transitional expressions, and so on) and coherence (narrative and sequential "logic," conventions for revealing hierarchical and categorical relationships, and so on);
- Differing uses of linguistic structures and devices (e.g., less subordination, more conjunction, less passivization, less noun modification, narrower vocabulary range, greater use of high-frequency vocabulary items, predictable variation in grammatical structure, simpler style, and so on).

(Sources: Belcher & Nelson, 2013; Connor, 2011; Connor et al., 2008; Ene et al., 2019; Hedgcock, 2012; Hinkel, 2002, 2005; Schlepppegrell & Christie, 2018)

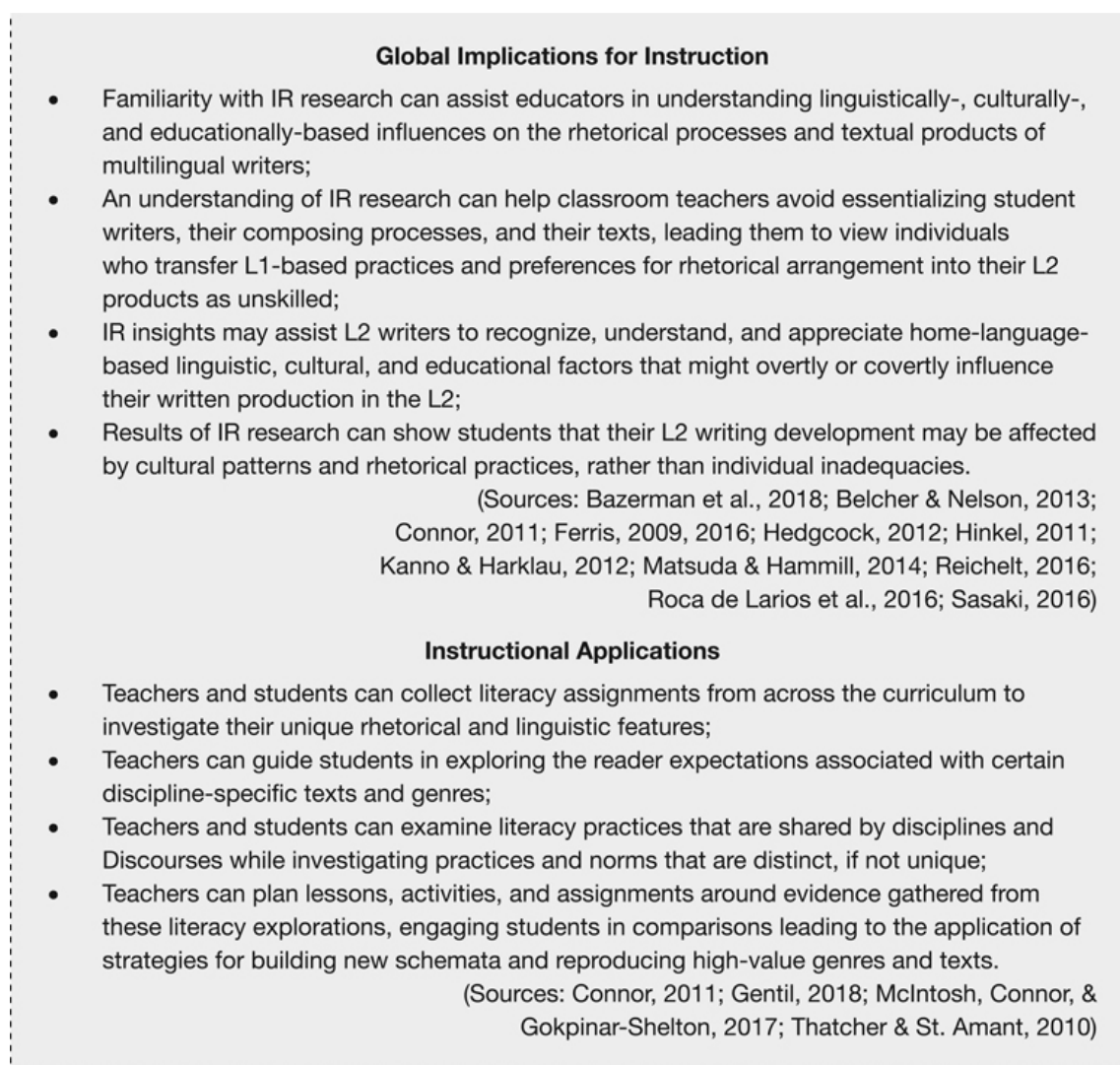


Figure 1.6 Principal differences between L1 and L2 student writing: Implications and applications.

Prior knowledge about texts, their genre categories, their purposes, and their formal properties make up learners' *schematic knowledge*, an individual's "broad knowledge about the world" (Willingham, 2017, p. 135). A *schema* consists of a script, frame, or mental framework that organizes our prior knowledge about a topic, text, or experience (Bartlett, 1932; Rumelhart, 1980; Rumelhart & McClelland, 1982; Rumelhart et al., 1986). People construct *schemata* (the plural form of *schema*) "for everything in their lives, including people, places, things, language, processes, and skills" (Tracey & Morrow, 2017, p. 60). According to *schema theory*, a person's expectations and assumptions about a text or speech event are "externally constructed and impose external constraints on the ways in which we understand messages" (Schiffrin, 1994, p. 104). Successful comprehension of texts can depend wholly or entirely on schemata, "which help us make sense of new facts, text types, formal patterns, and practices" (Hedgcock & Ferris, 2018, p. 39). In addition to consisting of organized *chunks* (clusters) of knowledge and experience, a schema may also trigger expectations, attitudes, and feelings. In aggregate, schemata "explain the constitutive role of culturally organized experience in individual sense making" (McVee et al., 2005, p. 535).

Researchers have classified schematic knowledge into several subcategories, three of which we will describe here. The first of these is usually characterized as *content schemata*, an individual's pre-existing familiarity with the ideas expressed in an oral or written text. Content schemata are

naturally crucial for both readers and writers of any language. They are especially vital for L2 readers and writers, as incomplete content knowledge can create serious comprehension gaps when students encounter unfamiliar texts ([Alptekin & Erçetin, 2011](#); [Grabe & Stoller, 2020](#); [Hedgcock & Ferris, 2018](#); [Keshavarz et al., 2007](#)). *Cultural schemata*, which some theorists have called *abstract schemata*, include awareness of culture-specific practices, traditions, relationships, identities, beliefs, and values ([Ketchum, 2006](#); [Nassaji, 2002](#)). We develop cultural schemata in our everyday experiences as human beings – in the course of routine encounters with others and with culturally embedded forms of expression ([Alptekin, 2006](#); [Kern, 2000](#)). [Ketchum \(2006\)](#) characterized cultural schemata as a kind of extension of content schemata, as the two are often inextricably intertwined. That is, in order to understand a text or participate in a speech event, a person must have established a role (however emergent) as a member of the culture. A third category, *formal schemata*, entails what individuals know about the rhetorical structure of texts and speech events ([Smith, 2004](#)). A related category, *linguistic schemata*, includes knowledge of the morphosyntactic properties and lexical choices associated with particular genres and speech events, as well as with particular topical areas ([Toledo, 2005](#)).⁷

Investigations of the influences of schematic knowledge on language learning, L2 reading development, and L2 reading efficiency consistently reveal that extensive and sophisticated schematic knowledge measurably facilitates text comprehension (and literacy skills more broadly). Conversely, the absence of sufficient schematic knowledge can inhibit comprehension and even debilitate novice readers and writers.⁸ Literacy events such as reading or writing a text fundamentally involve an interaction between readers and texts, and readers' responses to a text vary according to their schemata and other circumstances at the time of the interaction. To illustrate how our multiple schemata assist us when we encounter an unfamiliar text, imagine that you are glancing over a recipe on a website or in a print source. What would you expect the content and organization of a recipe to consist of? You would probably anticipate finding the dish's title, a list of required ingredients and their measurements, and then a sequential list of steps to follow in order to prepare the dish. A recipe that deviates from this expected structure might confuse or even frustrate you. A recipe that provides exceedingly technical instructions or that uses low-frequency vocabulary to describe ingredients and steps might require you to pause and re-read. Your pre-existing expectations about the text's organization, substance, and linguistic features constitute your content, cultural, and formal schemata. Naturally, if you have never cooked a meal or baked a cake, your culinary schemata will be limited – and quite different from the schematic repertoire of a skilled cook. After all, your experiences in the kitchen differ from those of people who have spent more time cooking. In addition, your experience reading that recipe will vary depending on the circumstances and your purposes: Are you merely glancing at the text to consider it for a future meal? Are you in the kitchen at this moment preparing tonight's dinner? The specific situation will obviously affect your level of concentration and the attention to detail that you bring to the reading task.

A chief implication of schema theory and its research findings is that teachers must take systematic steps to ensure that learners find the topics, texts, and other learning materials of a course cognitively, developmentally, and culturally accessible. Teachers of writing can achieve these goals by selecting reading materials and composing tasks that engage learners in capitalizing on their prior experience, in activating their existing schemata, and in building this schematic knowledge. We offer specific strategies and procedures for schema activation and development in [Chapter 4](#).

Intercultural Rhetoric and Its Implications for Teaching L2 Writing

As [Figure 1.6](#) suggests, a significant way in which the schemata of L2 writers differ from those of L1 writers involves their expectations and assumptions about the rhetorical conventions and social functions of texts (see [Chapter 2](#)). The study of *intercultural rhetoric* (IR) aims to characterize these divergent assumptions and expectations. [Connor \(2011\)](#) broadly defined IR, formerly called *contrastive rhetoric* (CR), as “the study of written discourse between and among individuals with different cultural backgrounds,” noting that discourse entails “language use beyond the sentence as well as social and ideological assumptions ... associated with communication” (p. 2). IR thus endeavors to uncover the effects of these divergent expectations, as well as formal contrasts, on the literacy development of multilingual writers. [Connor \(1996\)](#) observed that the comparative study of rhetorical patterns across languages and cultures presupposes that, because “language and writing are cultural phenomena ... each language has rhetorical conventions unique to it” (p. 5).

The Genesis of Intercultural Rhetoric: Contrastive Rhetoric

A pioneering empirical study by [Kaplan \(1966\)](#) is considered a landmark work that ignited intense interest in CR and initiated a robust line of research ([Almuhailib, 2019](#); [Belcher & Nelson, 2013](#); [Connor, 1996, 2011](#); [Connor & Kaplan, 1987](#); [Connor et al., 2008](#); [Enkvist, 1997](#); [Kassabgy et al., 2004](#); [Thatcher & St. Amant, 2010](#)). Described by [Bloch \(2013\)](#) as “the single most important contribution to the discussion of second language writing,” Kaplan’s article synthesized the disparate fields of applied linguistics and rhetoric “into a new, highly generative area of inquiry” (p. 243). At the time that Kaplan’s article appeared, *contrastive analysis* (CA) and its applications were very much in vogue. In the CA approach, linguists and materials developers compared the grammatical structures and phonological properties of learners’ primary languages with those of the target languages that they were acquiring. It was commonly believed that CA could identify specific areas of L2 learning difficulty (e.g., in syntax, morphology, and speech production) and that direct instruction focusing on major contrastive features could facilitate language teaching and accelerate the learning process ([Ellis, 2015](#); [Gass, 2020](#); [Mitchell et al., 2019](#)).

[Kaplan’s \(1966\)](#) study featured a formal analysis of more than 600 paragraph-length writing samples produced by university ESL students representing a range of mother tongues. A systematic rhetorical and linguistic examination of the student texts led him to propose several generalizations regarding the relationship of ESL writers’ L1 rhetorical patterns and the salient characteristics of their expository writing in English. Kaplan illustrated these patterns in the form of simple line drawings, which later came to be known as “doodles.” Against the backdrop of CA, Kaplan suggested that the contrasting rhetorical patterns identified in the ESL writing samples could be associated with the largely non-linear meaning-making patterns and rhetorical “styles” prevalent in students’ primary languages. For example, he observed that Arabic-speaking learners made extensive use of coordination in their L2 writing samples, to a degree considered excessive by English-speaking academic readers. The texts produced by speakers of Asian languages such as Chinese and Japanese tended to circle around a topic or argument, rather than approaching it linearly (head-on) in presenting an argument. In contrast, the writing of speakers of European languages such as Spanish and French tended to digress in exposing a topic or argument, often

introducing extraneous material more frequently than did English-speaking academic writers. Kaplan's inferences about student writers' L1 rhetorical preferences and practices, we should note, are based on very brief texts produced in English.

In subsequent work that significantly advanced research methods in the CR tradition, [Hinds \(1984, 1987, 1990\)](#) analyzed journalistic and expository texts produced by accomplished writers of languages such as Japanese, Korean, Chinese, and Thai. Drawing on systematic formal analysis of writers' L1 texts, Hinds inferred that Japanese and Korean writers consistently delayed revealing their purposes for writing until the end of their texts, apparently adhering to an inductive (in contrast to deductive) rhetorical pattern. Hinds hypothesized that the texts generated by Japanese and Korean writers reflected an effort to convince readers of the validity of their arguments before confirming their theses, thus gaining their readers' sympathy. In contrast, Hinds found that Anglo-American essay writers generally pre-revealed their purposes or arguments quite early in their texts, adhering to a predominantly deductive rhetorical structure. Whereas this direct, explicit rhetorical strategy tends to be valued by Anglo-American readers and is often explicitly taught in composition instruction (see [Chapters 3 and 4](#)), Japanese and Korean writers tended to avoid this pattern. Hinds proposed that such overtness and directness could be perceived by Japanese and Korean readers as too direct and perhaps even offensive to a reader who expects writers to expose their reasoning more subtly. These descriptive contrasts provided "a window into cross-linguistically diverse approaches to structuring written text," suggesting diverse, culturally embedded means of expressing meaning ([Hedgcock, 2012](#), p. 225).

It is helpful to remember that [Kaplan's \(1966\)](#) conclusions were primarily text-based and descriptive in nature, rather than predictive. Grounded in linguistics, his research methods were textual and quantitatively oriented ([Connor, 2014](#)). Kaplan's insights into the diversity of rhetorical practices across cultures, as well as the more nuanced CR and IR investigations that followed, tell us a great deal about the social and cultural influences that shape how people write. No student writer comes to the L2 writing classroom as a *tabula rasa*. Rather, most L2 students begin developing L2 writing skills with literate histories that inevitably influence their readiness and awareness (see [Chapter 2](#)). The insights of CR and IR can help teachers understand their student writers, their existing rhetorical knowledge, and their potential needs as L2 writers. This understanding is essential to designing writing instruction that effectively addresses the observable needs and expectations of particular student populations ([Connor, 2011](#); [Ene et al., 2019](#); [Ferris, 2009, 2016](#); [Hinkel, 2015](#); [Hirvela & Belcher, 2021](#); [Liu & Du, 2018](#)).

Controversies in CR/IR, Advances in Research, and Their Influence on L2 Writing Instruction

[Kaplan's \(1966\)](#) trailblazing study ignited considerable interest in CR, whose successor (IR) has generated an impressive body of empirical research. [Hinkel \(2002\)](#) credited Kaplan with articulating the assumption that L2 student writers in North America require instruction in order to compose "academic texts that are congruent with Anglo-American rhetorical paradigms" (p. 6) – a now questionable perspective that has met with strong objections. Early on, theories of linguistic relativity, rhetoric, discourse analysis, genre analysis, and translation informed the CR agenda. Emphasizing difference rather than commonality, CR research focused chiefly on the written products of expert and novice writers, in contrast to writers themselves, their writing processes, and the sociocultural contexts for writing ([Belcher, 2014](#)). CR/IR research has predictably led to spirited debates about the real and perceived influences of L2 writers' primary

languages and literacies, knowledge of L1-specific rhetorical patterns, and educational experiences on how they write (Casanave, 2017; Gentil, 2018; Kubota, 1998, 2010). Some criticism of CR has been particularly severe and not always fair: For example, Kubota (2014) asserted that “the original model of CR is clearly obsolete” (p. 119). CR has admittedly been controversial, but we believe that its successor IR has substantially advanced the field. By accounting for prototypical formal features in written discourse (see Figure 1.6), IR has contributed a great deal to our understanding of rhetorical and lexicogrammatical patterns in written text across linguistic boundaries and genres (Connor, 1996, 2003, 2011, 2014; Connor et al., 2008).

In the years following the publication of Kaplan’s (1966) study, L2 educators derived several instructional implications from CR research. For example, reflecting “assimilationist tendencies” of the era, many practitioners believed that the CR hypothesis confirmed the need for L2 writers to undergo explicit instruction and modeling of L2 rhetorical patterns, primarily English (Belcher, 2014). Predictably, this view dovetailed conveniently with the current-traditional approaches to writing instruction that prevailed from the 1960s well into the 1980s (see Chapter 3). As the CR hypothesis underwent increasing scrutiny, however, questions about its validity and empirical premises surfaced. Some early interpretations of CR implied strong connections between writers’ cognitive processes and their written output. Scholars have challenged and even rejected these hypothesized links, further charging that CR and IR research exhibits ethnocentric (if not Anglo-centric) overtones and that contrastive hypotheses are culturally deterministic (Belcher, 2014; Canagarajah, 2013a, 2013b; Casanave, 2017; Connor, 2011; Kirkpatrick & Xu, 2012; Kubota, 2010, 2014; Panetta, 2001).

Critics have similarly charged that early CR studies tended to draw broad generalizations about culture-specific textual patterns and rhetorical practices, often informed solely by L2 students’ written products (Connor, 2004; Panetta, 2001). Some studies have produced outcomes that do not support CR’s predictions about (negative) interlingual transfer. Hirose (2003) and Kubota (1998), for example, reported empirical results demonstrating that Japanese students identified and successfully transferred deductive features from Japanese into their English writing. More recently, Kirkpatrick and Xu (2012) examined Chinese rhetorical practices over several millennia. They reported that some Chinese literate traditions feature rhetorical conventions that strongly resemble the deductive arrangements characteristic of “Western” rhetoric, directly contradicting some leading CR claims. Kirkpatrick and Xu acknowledged a preference for induction, “chain reasoning,” and “reasoning by analogy and historical precedent” in classical Chinese texts, but they also found extensive evidence of deductive organization in both classical and contemporary texts (p. 139). Emphasizing the complexity and dynamism of Chinese rhetoric, Kirkpatrick and Xu concluded that Chinese textual conventions “change for the same ... reasons and in the same ... ways as writing styles in other great literate cultures” (p. 4). The work of Kirkpatrick and Xu and others demonstrates that social, cultural, political, ideological, and economic factors can influence such change, as can intercultural contact and educational policy. Such is the case in 21st-century China, where interaction with the West is extensive and where EFL writing instruction is prevalent (Hu, 2019; You, 2010, 2018). Contemporary approaches to IR research encourage us to avoid the “simplistic cultural determinism” associated with early iterations of CR (Belcher, 2014).

Kubota (2010) raised a related objection to CR research – namely, that CR constructed “a binary between English and non-English languages.” Kubota’s sharp and provocative critique further held that this binary implied “superiority of English and inferiority of other languages” (p. 267), a claim that we do not support. Critics have also asserted that CR approaches tended to characterize L2 writers’ texts as static, to overlook the role played by multilingual writers’

audiences in the L2 composing process, and to represent multilingual writers and their home cultures as homogeneous (that is, to *essentialize* them) ([Baker, 2013](#); [Canagarajah, 2013a](#); [Casanave, 2012, 2017](#); [Hedgcock, 2012](#); [Kubota, 2014](#); [Kuteeva & Mauranten, 2014](#); [Tardy & Whittig, 2017](#)). IR theorists and researchers have acknowledged this range of criticism as legitimate. As [Connor \(2011\)](#) pointed out, some objections to CR/IR have “stemmed from a lack of understanding about what [IR] stands for today” (p. 13). Researchers and theorists working toward a reframing and realignment of the IR agenda as a genuinely *intercultural* enterprise have diversified the focal areas of empirical inquiry ([Belcher, 2014](#); [Belcher & Nelson, 2013](#); [Connor, 2004, 2011, 2014](#); [Connor et al., 2008](#); [Ene et al., 2019](#); [McIntosh et al., 2017](#); [Thatcher, 2011](#); [Thatcher & St. Amant, 2010](#)). Contemporary approaches to IR research have accordingly “expanded the knowledge base about the nature of ... written discourse, suggesting principles for guiding L2 student writers toward crafting their written products” in ways that satisfy diverse reader expectations ([Hedgcock, 2012](#), p. 226).

We can attribute this significant and welcome turn in IR to several significant developments in the field, including:

- Broad acceptance of the premise that writer populations are heterogeneous and that multilingual writers can exercise agency and autonomy as they produce texts for specific audiences and construct their scribal identities;
- A discernible shift away from assumptions about homogeneous, monocultural, prescriptive, and primarily anglophone norms and standards by which multicultural writers and texts are examined and evaluated;
- Efforts to redefine *rhetoric* beyond the Aristotelian tradition, to include the precept that rhetoric entails not only persuasion and the arrangement of textual elements but also cognition, emotion, and rhetorical situation;
- A stronger emphasis on intercultural communication, with its focus on how languages, cultures, and discourse communities interact, mutually influence one another, and form new hybrid discourses;
- A corresponding recognition of flexible translingual flows across cultural and textual boundaries, in addition to a stronger focus on multilingual writers’ translingual practices (e.g., translation, code-mixing, code-switching, and code-meshing);
- Enhanced interdisciplinary cooperation and collaboration among rhetoricians and text linguists who work in diverse traditions;
- Broader application of comparative rhetoric, which explores the rhetorical conventions and practices specific to Western and non-Western languages, cultures, and literary traditions;
- Wider adoption of cultural studies and its research methods, which feature rigorous qualitative approaches such as case studies, participant and nonparticipant observation, ethnography, and autobiographical methods;
- Increasingly sophisticated methods of text-based research such as those now commonplace in genre analysis and corpus linguistics, as well as novel approaches to the empirical study of writing processes;

- Explicit acknowledgment of the interdependence of the traditional “four skills” (listening, speaking, reading, and writing), grammatical awareness, vocabulary knowledge, pragmatics, and cultural competence.

(Sources: [Belcher, 2014](#); [Belcher & Nelson, 2013](#); [Casanave, 2017](#); [Connor, 2011](#); [Ene et al., 2019](#); [McIntosh et al., 2017](#))

As a result of new thinking and efforts to address persistent criticism, the strong predictive claims made by early CR research no longer typify the study of IR or its implications for teaching L2 writing ([Connor, 2011, 2014](#); [Gonerko-Frej, 2014](#)). State-of-the-art IR research does not presuppose that writers from a given linguistic or cultural background will experience the same challenges in learning a given L2 or in undertaking specific writing tasks. The insights that IR has supplied concerning texts, genres, writers, and writing processes nonetheless provide composition teachers with a rich knowledge base and with empirical evidence on which to base decisions about teaching, response, and assessment. Knowing how patterns in written language differ cross-culturally and cross-linguistically, for example, can help both teachers and students in two influential ways. First, IR underscores the value of primary-language knowledge and literacy as resources for learning ([Ene et al., 2019](#); [Gentil, 2018](#); [Thatcher, 2011](#); [Thatcher & St. Amant, 2010](#)). Second, “IR provides a useful lens for looking at how multilingual writers can traverse languages and cultures – big and small – in a variety of genres” ([McIntosh et al., 2017](#), p. 14). [Figure 1.6](#) presents a partial list of differences between L1 and L2 student writing that is informed by IR research, as well as practical implications and applications to the teaching of L2 writing.

We believe that L2 composition educators must become aware of the rhetorical expertise that their novice writers bring to the writing classroom. This expertise may include formal, content, and cultural schemata, which can include writers’ implicit and explicit knowledge about textual arrangement, genres, and their purposes. Teachers must also recognize that L2 writers’ primary languages, home cultures, and prior educational experiences do not predetermine their cognitive advantages or the potential challenges that they may face in comprehending and producing unfamiliar rhetorical patterns. L2 writers’ knowledge bases differ not only from those of monolingual writers of English, but also from those of other L2 writers. Each L2 writer should be viewed in individual terms, not as a prototype representing a set of essentializing collective norms or stereotypes. Individual differences across L2 writers affect their abilities to comprehend, analyze, and respond to the texts that they read, to function effectively in L2 literacy classrooms, and to construct original texts that fulfill the expectations of target language readers. As a result, writing teachers must consider the implications of schematic and rhetorical differences in selecting texts for reading, designing and assigning literacy tasks, planning and executing lessons, providing feedback to student writers, and devising and using assessment instruments.

CHAPTER SUMMARY

Teachers in any discipline must develop not only subject-matter knowledge, but also the technical expertise to connect classroom instruction to theory and research findings. Because writing itself constitutes the fundamental subject matter of L2 writing instruction, this chapter has endeavored to provide readers with foundational knowledge about the origins, evolution, and key constituents

of modern writing systems. Our exploration of writing as a relatively recent and truly ingenious human innovation suggests that writing systems, orthographies, and scripts as we know them today require their users to develop sophisticated cognitive and creative skills, which can only be cultivated through explicit teaching. Our limited survey of writing systems and the literacies that have evolved around them should help us appreciate the innovative and culturally-embedded qualities of writing. A basic understanding of writing systems should also enable us to recognize writing as an invaluable means of communication and information storage, as well as socialization, learning, and creative expression. Although speech and writing share certain universal properties, oral language and writing systems vary considerably in terms of their structure and constituents. This variation across languages, writing systems, and cultures naturally necessitates language and literacy education that builds on learners' existing knowledge; L2 instruction must also systematically target key features of the new language and its literate practices. For these reasons, we believe that the findings and insights of IR research can helpfully inform the design and delivery of L2 writing instruction.

Since its inception, the field of L2 writing has made considerable progress in its pursuit of robust theory and research geared toward equipping classroom teachers with practical tools for planning and implementing effective instruction for diverse and ever-changing populations of multilingual writers. Indeed, the discipline has made impressive strides since the latter decades of the 20th century, when L2 composition instruction took divergent (and even contradictory) forms, lacked methodological consensus, and was only weakly guided by a coherent theory of literacy development. Researchers in pursuit of a definitive and comprehensive understanding of optimal methods for cultivating multilingual learners' writing skills have made significant strides since the 1980s and 1990s, thanks largely to advances in IR. As subsequent chapters of this volume reveal, L2 writing research has shed light on the distinct backgrounds and needs of multilingual writers. This increasingly nuanced awareness of L2 writers and their trajectories has contributed to a complex view that accounts for the cognitive, linguistic, social, cultural, educational, and affective factors that distinguish multilingual writers, L2 writing processes, L2 texts, and contexts for writing. Extensive research on writing in diverse socioeducational contexts has led us to an increasingly sophisticated understanding of student writers, texts, processes, and contexts. This work, surveyed in the remainder of this book, can assist educators in making informed decisions about instructional planning, teaching practice, writing assessment, and approaches to feedback.

FURTHER READING AND RESOURCES

As in subsequent chapters, we present a list of selected materials that readers may consult for more in-depth treatments of topics introduced in this chapter. Complete information for published sources can be found in the Bibliography at the end of this chapter. We provide selected web addresses for online resources, though we caution readers that URLs are subject to change.

- Intercultural rhetoric: [Belcher and Nelson \(2013\)](#); [Connor \(2011\)](#); [Connor et al. \(2008\)](#); [Eddy and Espinosa-Aquilar \(2019\)](#); [Thatcher \(2011\)](#)
- L1 literacy, writing development, and composition studies: [Bazerman et al. \(2018\)](#)
- L2 writing studies: [Manchón and Matsuda \(2016\)](#); [Manchón and Polio \(2022\)](#)
- Writing systems and their evolution: [Baron \(2009\)](#); [Cook and Ryan \(2020\)](#); [Daniels and Bright \(1996\)](#); [Grigorenko et al. \(2012\)](#); [Rogers \(2004\)](#); [Seidenberg \(2017a\)](#)

- Omniglot Online Encyclopedia of Writing Systems: www.omniglot.com

REFLECTION AND REVIEW

1. Identify and reflect on the roles of formal theory in teacher learning and instructional practice. How can systematic knowledge about theoretical models and research traditions meaningfully contribute to one's practices and skills as a classroom teacher of writing?
2. Why do you think scholars consider the introduction of writing to be such an influential sociohistorical innovation? In what ways does writing constitute a novel technology?
3. In your view, what are the most significant ways in which speech and writing are similar and different? Why is it important for teachers of writing to appreciate the unique properties of writing systems and processes?
4. Considering the unique characteristics of the various categories of writing systems surveyed in this chapter, what are the potential advantages and disadvantages of each for communication, learning, and teaching?
5. Review the categories of schematic knowledge, reflecting on how they might help (or possibly hinder) novice L2 writers in reading, processing, and generating L2 texts. Which schemata do you believe are particularly crucial for your L2 writers to develop?
6. In what ways might the study of intercultural rhetoric (IR) inform the planning and delivery of L2 composition courses? How might IR observations about culturally embedded rhetorical patterns influence the selection of texts and the development of assignments in a writing course? Identify IR implications for assessing and responding to student writing.

APPLICATION ACTIVITIES

Application Activity 1.1 Exploring Writing Systems

Individually or with a fellow student, examine and compare two or more writing systems and summarize your main discoveries in a short written report or brief presentation. Follow the steps and prompts outlined here:

1. Visit the Omniglot website: www.omniglot.com.
2. On the main page, select *Writing Systems*, and then choose at least two different classes of writing system. Make sure that one of these writing systems is unlike any script that you already know. For instance, you might select an alphabet and a syllabary, or an abugida (alphasyllabary) and a logographic (semanto-phonetic) writing system.
3. To facilitate your analysis and comparison, create a simple comparison table. In exploring and learning about each script, note the following features:
 - a. Script classification and relative transparency
 - b. Relationship between script and speech forms

- c. Historical origins
 - d. Direction of writing
 - e. Number and structure of graphemes
 - f. Languages written in the script
 - g. General rules for writing in the script (e.g., some abjads require pointing of vowels; logographic writing may prescribe stroke sequences)
4. Reflect on what your comparison reveals about the scripts' phonographic characteristics (i.e., how the scripts capture oral language and represent meaning in distinct ways), noting particularly novel or surprising insights.
 5. Discuss the potential cognitive difficulties of learning a new writing system, script, and orthography: How might your understanding of writing systems inform your practice as a teacher of L2 writing?

Application Activity 1.2 Schematic and Rhetorical Text Analysis

For this exercise, select a self-contained authentic text of 750 to 1,500 words that you might use as a reading passage in an L2 writing course, such as: a selection from a course reader or anthology, a literary passage, a nonacademic informational text for a general audience, a news article from a reputable news source, or an op-ed (See [Appendix 4.2](#) for a more expansive list of genres to choose from). Analyze the selection by addressing the following prompts. Report and interpret your findings in a written report or brief presentation.

1. What content, cultural, and formal schemata does the author of the text likely assume on the part of the reader? Identify features of the text that you think might be culturally embedded and worthy of explicit instruction. How might you engage your students in comparing textual content and structure to the content and structure of similar texts in their primary language(s)?
2. Considering what you know about your prospective students' existing schematic knowledge, what steps would you take to bridge their potential knowledge gaps and equip them to understand the text successfully? For example, what kinds of assistance with topical knowledge, rhetorical arrangement, challenging grammatical constructions, and unfamiliar vocabulary would you provide?
3. Devise a simple lesson outline, along with appropriate materials for assisting students with predicting the text's contents and structure, understanding and interpreting it, and ultimately responding to it in a writing task or exercise (e.g., a reading journal entry, a critical summary, a reaction statement posted as a blog entry).

NOTES

1. [The](#) evolving and controversial ways in which second language (L2) learners and writers have been characterized makes it difficult to arrive at an unproblematic yet meaningful way to describe multilingual students of English around the world. Chapter 2 addresses these challenges, providing an account of diverse learner populations and the contexts where they may learn to write (see [Tardy & Whittig, 2017](#)). Throughout this book, we use terms such as second language (L2), English as a second

language (ESL), and English as an additional language (EAL) to refer inclusively to settings in which English is taught as a second or additional language (i.e., in contexts where English is a dominant language, such as North America, the UK, Australia, New Zealand, and the rest of the British Commonwealth), as a foreign language (i.e., in non-Anglophone contexts where English is not a language of public life and where access to English outside the classroom is limited, such as South America, Europe, and regions of Africa and Asia), and as an additional language (i.e., in plurilingual contexts such as India, Hong Kong, Singapore, the Philippines, and South Africa, for instance). Though not optimal, the ESL designation to some extent describes the status of English in numerous sociolinguistic contexts and represents the diverse forms of multilingualism exhibited by many novice writers of English. The ESL initialism also allows us to avoid awkward conventions such as “ESL/EFL.”

2. [Here](#) and throughout this chapter, we use the following linguistic conventions to distinguish specific kinds of representation: (a) square brackets [] signify *phonetic* (spoken) representation; (b) forward slashes / / signify *phonemic* (underlying) representation; (c) chevrons (angled brackets) < > signify orthographic representation (conventional spelling).
3. [Somewhat](#) controversially, [Powell \(2012\)](#) argued that “even in China the idea of ‘writing’ must have come from Mesopotamia over the Gansu corridor north of the Himalayas, where caravan traffic was constant. China was never wholly separated from cultural developments in Mesopotamia” (p. 4).
4. [Chinese](#) writing is not purely logographic, even though it is persistently classified as such ([Tsu, 2022](#)). As [Seidenberg \(2017a\)](#) insisted, Chinese writing “is not logographic because it does not employ a separate sign for **every** word,” a characteristic that would place extraordinary demands on human memory (emphasis added). The Chinese system unquestionably includes some logographs, “but they are a small minority” (p. 47).
5. [Olson \(2012\)](#) and other critics (e.g., [Cole & Cole, 2006](#); [Daniels, 2009](#); [Powell, 2012](#)) have challenged this conventional view on the grounds that writing (or at least protowriting) predates Greek and Roman civilizations and that such systems emerged independently in far-flung geographic locations. [Olson \(2012\)](#) also cautioned against the ethnocentric tendency in Western scholarship to equate the discovery of the alphabetic principle with “literate society, and more grandly, civilization, [and] the society of rules and laws” (p. 37).
6. [Research](#) on the differential impact of L1 orthographic knowledge on L2 literacy learning is unfortunately too extensive to synthesize here. Helpful overviews of relevant research include: [Birch and Fulop \(2021\)](#); [Grabe \(2009\)](#); [Grabe and Stoller \(2020\)](#); [Hedgcock and Ferris \(2018\)](#); [Joshi and Aaron \(2006\)](#); [Koda \(2005, 2011\)](#); and [Koda and Zehler \(2008\)](#).
7. [Classifications](#) of schematic knowledge are not uniform: *Linguistic schema* sometimes refers to what we have described here as *cultural schema* (see: [Erten & Razi, 2009](#); [Ketchum, 2006](#)). In this volume, we use the term *linguistic schemata* to refer to knowledge about vocabulary (which entails the writing system and spelling) and grammar (which entails syntax and morphology).
8. [For](#) overviews of the impact of schematic knowledge on L2 reading, see: [Bernhardt \(2010\)](#); [Chen et al. \(2016\)](#); [Grabe \(2009\)](#); [Grabe and Stoller \(2020\)](#); [Hedgcock and Ferris \(2018\)](#); [Koda \(2005, 2011\)](#); and [Koda and Zehler \(2008\)](#).

BIBLIOGRAPHY

- [Ager, S.](#) (2021). *Sumerian* [Webpage]. www.omniglot.com/writing/sumerian.htm
- [Almuhailib, B.](#) (2019). Analyzing cross-cultural writing differences using contrastive rhetoric: A critical review. *Advances in Language and Literary Studies*, 10(2), 102–106.
- [Alptekin, C.](#) (2006). Cultural familiarity in inferential and literal comprehension in L2 reading. *System*, 34, 494–508.
- [Alptekin, C., & Erçetin, G.](#) (2011). Effects of working memory capacity and content familiarity on literal and inferential comprehension in L2 reading. *TESOL Quarterly*, 45, 235–266.
- [Atkinson, D.](#) (2010). Between theory with a big T and practice with a small p. In T. Silva & P. K. Matsuda (Eds.), *Practicing theory in second language writing* (pp. 5–18). West Lafayette, IN: Parlor Press.
- [Baker, W.](#) (2013). Interpreting the culture in intercultural rhetoric: A critical perspective from English as a lingua franca studies. In D. Belcher & G. Nelson (Eds.), *Critical and corpus-based approaches to intercultural rhetoric* (pp. 22–45). Ann Arbor, MI: University of Michigan Press.
- [Baron, D.](#) (2009). *A better pencil: Readers, writers, and the digital revolution*. Oxford, England: Oxford University Press.
- [Bartlett, C.](#) (1932). *Remembering*. Cambridge, England: Cambridge University Press.
- [Barton, D., & Lee, C.](#) (2013). *Language online: Investigating digital texts and practices*. New York, NY: Routledge.
- [Bauer, T.](#) (1996). Arabic writing. In P. Daniels & W. Bright (Eds.), *The world's writing systems* (pp. 559–564). Oxford, England: Oxford University Press.
- [Bazerman, C., Applebee, A. N., Berninger, V. W., Brandt, D., Graham, S., Jeffery, J. V., Matsuda, P. K., Murphy, S., Rowe, D. W., Schleppegrell, M., & Wilcox, K. C.](#) (2018). *The lifespan development of writing*. Urbana, IL: National Council of Teachers of English. <https://wac.colostate.edu/books/ncte/lifespan-writing/>
- [Bazerman, C., Graham, S., Applebee, A., Matsuda, P., Berninger, V., Murphy, S., Brandt, D. W., & Schleppegrell, M.](#) (2017). Taking the long view on writing development. *Research in the Teaching of English*, 51(3), 351–360. www.jstor.org/stable/44821267
- [Belcher, D.](#) (2012). Considering what we know and need to know about second language writing. *Applied Linguistics Review*, 3(1), 131–150.
- [Belcher, D.](#) (2014). What we need and don't need intercultural rhetoric for: A retrospective and prospective look at an evolving research area. *Journal of Second Language Writing*, 25, 59–67.
- [Belcher, D., & Nelson, G.](#) (Eds.). (2013). *Critical and corpus-based approaches to intercultural rhetoric*. Ann Arbor, MI: University of Michigan Press.
- [Bernhardt, E. B.](#) (2010). *Understanding advanced second-language reading*. New York, NY: Routledge.
- [Berninger, V. W., & Chanquoy, L.](#) (2012). What writing is and how it changes across early and middle childhood development: A multidisciplinary perspective. In E. L. Grigorenko, E. Mambrino, & D. D. Preiss (Eds.), *Handbook of writing: A mosaic of perspectives and views* (pp. 65–84). New York, NY: Psychology Press.
- [Biber, D.](#) (2009). Are there linguistic consequences of literacy? Comparing the potentials of language use in speech and writing. In D. R. Olson & N. Torrance (Eds.), *Cambridge handbook of literacy* (pp. 75–91). New York, NY: Cambridge University Press.

- [Birch, B., & Fulop, S.](#) (2021). *English L2 reading: Getting to the bottom* (4th ed.). New York, NY: Routledge.
- [Bloch, J.](#) (2013). Afterword. In D. Belcher & G. Nelson (Eds.), *Critical and corpus-based approaches to intercultural rhetoric* (pp. 243–252). Ann Arbor, MI: University of Michigan Press.
- [Borg, S.](#) (2009). Language teacher cognition. In A. Burns & J. C. Richards (Eds.), *The Cambridge guide to second language teacher education* (pp. 144–152). Cambridge, England: Cambridge University Press.
- [Brown, A.](#) (2019). *Understanding and teaching English spelling*. New York, NY: Routledge.
- [Canagarajah, A. S.](#) (2013a). From intercultural rhetoric to cosmopolitan practice: Addressing new challenges in lingua franca English. In D. Belcher & G. Nelson (Eds.), *Critical and corpus-based approaches to intercultural rhetoric* (pp. 203–226). Ann Arbor, MI: University of Michigan Press.
- [Canagarajah, A. S.](#) (2013b). *Translingual practice: Global Englishes and cosmopolitan relations*. New York, NY: Routledge.
- [Casanave, C. P.](#) (2012). Controversy and change in how we view L2 writing in international contexts. In L. Alsagoff, S. L. McKay, G. Hu, & W. A. Renandya (Eds.), *Principles and practices for teaching English as an international language* (pp. 282–298). New York, NY: Routledge.
- [Casanave, C. P.](#) (2017). *Controversies in second language writing: Dilemmas and decisions in research and instruction* (2nd ed.). Ann Arbor, MI: University of Michigan Press.
- [Chen, X., Dronjic, V., & Helms-Park, R.](#) (Eds.). (2016). *Reading in a second language: Cognitive and psycholinguistic issues*. New York, NY: Routledge.
- [Cole, M., & Cole, J.](#) (2006). Rethinking the Goody myth. In D. R. Olson & M. Cole (Eds.), *Technology, literacy, and the evolution of society: Implications of the work of Jack Goody* (pp. 305–324). Mahwah, NJ: Lawrence Erlbaum.
- [Connor, U.](#) (1996). *Contrastive rhetoric: Cross-cultural aspects of second-language writing*. Cambridge, England: Cambridge University Press.
- [Connor, U.](#) (2003). Changing currents in contrastive rhetoric: Implications for teaching and research. In B. Kroll (Ed.), *Exploring the dynamics of second language writing* (pp. 218–241). Cambridge, England: Cambridge University Press.
- [Connor, U.](#) (2004). Intercultural rhetoric research: Beyond texts. *Journal of English for Academic Purposes*, 4, 291–304.
- [Connor, U.](#) (2011). *Intercultural rhetoric in the writing classroom*. Ann Arbor, MI: University of Michigan Press.
- [Connor, U.](#) (2014). Comments on Xiaoming Li's 'Are "cultural differences a mere fiction"?'. *Journal of Second Language Writing*, 25, 121–122. <http://dx.doi.org/10.1016/j.jslw.2014.06.009>
- [Connor, U., & Kaplan, R. B.](#) (Eds.). (1987). *Writing across languages: Analysis of L2 text*. Reading, MA: Addison-Wesley.
- [Connor, U., Nagelhout, E., & Rozycki, W. V.](#) (Eds.). (2008). *Contrastive rhetoric: Reaching to intercultural rhetoric*. Amsterdam, The Netherlands: John Benjamins.
- [Cook, V. J.](#) (2004). *The English writing system*. London, England: Edward Arnold.

- [Cook, V. J., & Bassetti, B.](#) (Eds.). (2005). *Second language writing systems*. Clevedon, England: Multilingual Matters.
- [Cook, V. J., & Ryan, D.](#) (Eds.). (2020). *Routledge handbook of the English writing system*. New York, NY: Routledge.
- [Coulmas, F.](#) (1996). *The Blackwell encyclopedia of writing systems*. Malden, MA: Blackwell.
- [Crystal, D.](#) (2006). *How language works: How babies babble, words change meaning, and languages live or die*. Woodstock, NY: Overlook.
- [Crystal, D.](#) (2011). *Internet linguistics: A student guide*. New York, NY: Routledge.
- [Cumming, A.](#) (2010). Theories, frameworks, and heuristics: Some reflections on inquiry and second language writing. In T. Silva & P. K. Matsuda (Eds.), *Practicing theory in second language writing* (pp. 19–47). West Lafayette, IN: Parlor Press.
- [Danet, B.](#) (2001). *Cyberpl@y: Communication online*. Oxford, England: Berg.
- [Daniels, P. T.](#) (2009). Grammatology. In D. R. Olson & N. Torrance (Eds.), *Cambridge handbook of literacy* (pp. 25–45). New York, NY: Cambridge University Press.
- [Daniels, P. T., & Bright, W.](#) (Eds.). (1996). *The world's writing systems*. New York, NY: Oxford University Press.
- [Defior, S., Cary, L., & Martos, F.](#) (2002). Differences in reading acquisition development in two shallow orthographies: Portuguese and Spanish. *Applied Psycholinguistics*, 23, 135–148.
- [DeFrancis, J.](#) (1989). *Visible speech: The diverse oneness of writing systems*. Honolulu, HI: University of Hawai'i Press.
- [Dehaene, S.](#) (2009). *Reading in the brain: The new science of how we read*. New York, NY: Penguin.
- [Diringer, D.](#) (1968). *The alphabet: A key to the history of mankind* (3rd ed.). London, England: Hutchinson.
- [Eddy, R., & Espinosa-Aquilar, A.](#) (Eds.). (2019). *Writing across cultures*. Ogden, UT: Utah State University Press.
- [Ellis, R.](#) (2015). *Understanding second language acquisition* (2nd ed.). Oxford, England: Oxford University Press.
- [Ene, E., McIntosh, K., & Connor, U.](#) (2019). Using intercultural rhetoric to examine translingual practices of postgraduate L2 writers of English. *Journal of Second Language Writing*, 45, 105–110.
- [Enkvist, N. E.](#) (1997). Why we need contrastive rhetoric. *Alternation*, 4, 188–206.
- [Erten, I. H., & Razi, S.](#) (2009). The effects of cultural familiarity on reading comprehension. *Reading in a Foreign Language*, 21, 60–77.
- [Ferris, D. R.](#) (2009). *Teaching college writing to diverse student populations*. Ann Arbor, MI: University of Michigan Press.
- [Ferris, D. R.](#) (2016). L2 writers in higher education. In R. Manchón & P. K. Matsuda (Eds.), *Handbook of second and foreign language writing* (pp. 141–160). Berlin, Germany: de Gruyter Mouton.
- [Finnegan, R.](#) (1988). *Literacy and orality: Studies in the technology of communication*. Oxford, England: Blackwell.

- [Freeman, D., Webre, A.-C., & Epperson, M.](#) (2019). What counts as knowledge in English language teaching? In S. Walsh & S. Mann (Eds.), *Routledge handbook of English language teacher education* (pp. 13–24). New York, NY: Routledge.
- [Gass, S., with Behney, J., & Plonsky, L.](#) (2020). *Second language acquisition: An introductory course* (5th ed.). New York, NY: Routledge.
- [Gee, J. P.](#) (2015). *Social linguistics and literacies: Ideology in discourses* (5th ed.). New York, NY: Routledge.
- [Gentil, G.](#) (2018). Multilingualism as a writing resource. In J. I. Lontas (Ed.), *The TESOL encyclopedia of English language teaching* (pp. 1–7). Hoboken, NH: John Wiley & Sons. doi: [10.1002/9781118784235.eelt0537](https://doi.org/10.1002/9781118784235.eelt0537)
- [Gonerko-Frej, A.](#) (2014). Teaching academic writing for the global world in Poland: The ELF perspective. In K. Bennett (Ed.), *The semi-periphery of academic writing* (pp. 75–90). New York, NY: Palgrave Macmillan.
- [Grabe, W.](#) (2009). *Reading in a second language: Moving from theory to practice*. New York, NY: Cambridge University Press.
- [Grabe, W., & Stoller, F. L.](#) (2020). *Teaching and researching reading* (3rd ed.). New York, NY: Routledge.
- [Grigorenko, E., Mambrino, E., & Preiss, D.](#) (Eds.). (2012). *Handbook of writing: A mosaic of perspectives and views*. New York, NY: Psychology Press.
- [Hedgcock, J. S.](#) (2002). Toward a socioliterate approach to language teacher education. *Modern Language Journal*, 86, 299–317.
- [Hedgcock, J. S.](#) (2009). Acquiring knowledge of discourse conventions in teacher education. In A. Burns & J. C. Richards (Eds.), *The Cambridge guide to second language teacher education* (pp. 144–152). Cambridge, England: Cambridge University Press.
- [Hedgcock, J. S.](#) (2010). Theory-and-practice and other questionable dualisms in L2 writing. In T. Silva & P. K. Matsuda (Eds.), *Practicing theory in second language writing* (pp. 229–244). West Lafayette, IN: Parlor Press.
- [Hedgcock, J. S.](#) (2012). Second language writing processes among adolescent and adult learners. In E. L. Grigorenko, E. Mambrino, & D. D. Preiss (Eds.), *Handbook of writing: A mosaic of perspectives and views* (pp. 219–237). New York, NY: Psychology Press.
- [Hedgcock, J. S., & Ferris, D. R.](#) (2018). *Teaching readers of English: Students, texts, and contexts* (2nd ed.). New York, NY: Routledge.
- [Hill, W.](#) (2020). Typography and the printed English text. In V. Cook & D. Ryan (Eds.), *Routledge handbook of the English writing system* (pp. 431–450). New York, NY: Routledge.
- [Hinds, J.](#) (1984). Retention of information using a Japanese style of presentation. *Studies in Language*, 8, 45–69.
- [Hinds, J.](#) (1987). Reader vs. writer responsibility: A new typology. In U. Connor & R. B. Kaplan (Eds.), *Writing across languages: Analysis of L2 text* (pp. 141–152). Reading, MA: Addison-Wesley.
- [Hinds, J.](#) (1990). Inductive, deductive, quasi-inductive: Expository writing in Japanese, Korean, Chinese, and Thai. In U. Connor & A. Johns (Eds.), *Coherence in writing* (pp. 87–110). Alexandria, VA: TESOL.

- [Hinkel, E.](#) (2002). *Second language writers' text: Linguistic and rhetorical features*. Mahwah, NJ: Lawrence Erlbaum Associates.
- Hinkel, E. (2005). Analyses of L2 text and what can be learned from them. In E. Hinkel (Ed.), *Handbook of research in second language teaching and learning* (pp. 615–628). Mahwah, NJ: Lawrence Erlbaum.
- [Hinkel, E.](#) (2015). *Effective curriculum for teaching L2 writing: Principles and techniques*. New York, NY: Routledge.
- [Hirose, K.](#) (2003). Comparing L1 and L2 organizational patterns in the argumentative writing of Japanese EFL students. *Journal of Second Language Writing*, 3, 203–181–209.
- [Hirvela, A., & Belcher, D.](#) (Eds.). (2021). *Argumentative writing in a second language: Perspectives on research and pedagogy*. Ann Arbor, MI: University of Michigan Press.
- [Holmes, J., & Wilson, N.](#) (2017). *An introduction to sociolinguistics* (5th ed.). New York, NY: Routledge.
- [Hu, Y.](#) (2019). Intertextualities in English writing of EFL learners in the context of a Chinese university. *Journal of Language Teaching and Research*, 10(2), 271–280.
- [Hyland, K.](#) (2019). *Second language writing* (2nd ed.). Cambridge, England: Cambridge University Press.
- [Janson, T.](#) (2012). *The history of languages: An introduction*. Oxford, England: Oxford University Press.
- [Joshi, R. M., & Aaron, P. G.](#) (Eds.). (2006). *Handbook of orthography and literacy*. Mahwah, NJ: Lawrence Erlbaum.
- [Kanno, Y., & Harklau, L.](#) (Eds.). (2012). *Linguistic minority students go to college: Preparation, access, and persistence*. New York, NY: Routledge.
- [Kaplan, R. B.](#) (1966). Cultural thought patterns in intercultural communication. *Language Learning*, 16, 1–20.
- [Kassabgy, N., Ibrahim, Z., & Aydelott, S.](#) (2004). *Contrastive rhetoric: Issues, insights, and pedagogy*. Cairo, Egypt: University of Cairo Press.
- [Katz, L., & Frost, R.](#) (1992). Reading in different orthographies: The orthographic depth hypothesis. In L. Katz & R. Frost (Eds.), *Orthography, phonology, morphology, and meaning* (pp. 67–84). Amsterdam, The Netherlands: Elsevier.
- [Kern, R.](#) (2000). *Literacy and language teaching*. Oxford, England: Oxford University Press.
- [Keshavarz, M. H., Atai, M. R., & Ahmadi, H.](#) (2007). Content schemata, linguistic simplification, and EFL readers' comprehension and recall. *Reading in a Foreign Language*.
- [Ketchum, E. M.](#) (2006). The cultural baggage of second language reading: An approach to understanding. *Foreign Language Annals*, 39, 22–42.
- [Kirkpatrick, A., & Xu, Z.](#) (2012). *Chinese rhetoric and writing: An introduction for language teachers*. Anderson, SC: Parlor Press.
- [Koda, K.](#) (2005). *Insights into second language reading: A cross-linguistic approach*. Cambridge, England: Cambridge University Press.
- [Koda, K.](#) (2011). Learning to read in new writing systems. In M. H. Long & C. J. Doughty (Eds.), *Handbook of language teaching* (pp. 463–485). Malden, MA: Wiley-Blackwell.

- [Koda, K., & Zehler, A.](#) (Eds.). (2008). *Learning to read across languages: Cross-linguistic relationships in first- and second-language literacy development*. London, England: Routledge.
- [Kubota, R.](#) (1998). An investigation of L1-L2 transfer in writing among Japanese university students: Implications for contrastive rhetoric. *Journal of Second Language Writing*, 7, 69–100.
- [Kubota, R.](#) (2010). Critical approaches to theory in second language writing: A case of critical contrastive rhetoric. In T. Silva & P. K. Matsuda (Eds.), *Practicing theory in second language writing* (pp. 191–208). West Lafayette, IN: Parlor Press.
- [Kubota, R.](#) (2014). Conceptual confusions and contradictions: A response to Professor Xiaoming Li. *Journal of Second Language Writing*, 25, 118–120. <http://dx.doi.org/10.1016/j.jslw.2014.06.008>
- [Kuteeva, M., & Mauranen, A.](#) (2014). Writing for publication in multilingual contexts: An introduction to the special issue. *Journal of English for Academic Purposes*, 13, 1–4. <http://dx.doi.org/10.1016/j.jeap.2013.11.002>
- [Larsen, D.](#) (2013). Focus on pre-service preparation for ESL writing instruction: Secondary teacher perspectives. In L. C. de Oliveira & T. Silva (Eds.), *L2 writing in secondary classrooms: Student experiences, academic issues, and teacher education* (pp. 119–132). New York, NY: Routledge.
- [Leong, C. K.](#) (1995). Orthographic and psycholinguistic considerations in developing literacy in Chinese. In I. Taylor & D. R. Olson (Eds.), *Scripts and literacy: Reading and learning to read alphabets, syllabaries, and characters* (pp. 163–183). Dordrecht, The Netherlands: Kluwer Academic.
- [Lewin, K.](#) (1951). *Field theory in social science*. New York, NY: Harper Torchbooks.
- [Liu, Y., & Du, Q.](#) (2018). Intercultural rhetoric through a learner lens: American students' perceptions of evidence use in Chinese yìlùnwén writing. *Journal of Second Language Writing*, 40, 1–11.
- [Lock, A., & Gers, M.](#) (2012). The cultural evolution of written language and its effects. In E. L. Grigorenko, E. Mambrino, & D. D. D. Preiss (Eds.), *Handbook of writing: A mosaic of perspectives and views* (pp. 11–35). New York, NY: Psychology Press.
- [Mair, V.](#) (1996). Modern Chinese writing. In P. Daniels & W. Bright (Eds.), *The world's writing systems* (pp. 200–208). Oxford, England: Oxford University Press.
- [Manchón, R. M., & Matsuda, P. K.](#) (Eds.). (2016). *Handbook of second and foreign language writing*. Berlin, Germany: de Gruyter Mouton.
- [Manchón, R. M., & Polio, C.](#) (Eds.). (2022). *Handbook of second language acquisition and writing* (2nd ed.). Abingdon, England: Routledge.
- Matsuda, P. K., & Hammill, M. J. (2014). Second language writing pedagogy. In G. Tate, A. R. Taggart, K. Schick, & H. B. Hessler (Eds.), *A guide to composition pedagogies* (2nd ed., pp. 266–282). New York, NY: Oxford University Press.
- [McIntosh, K., Connor, U., & Gokpinar-Shelton, E.](#) (2017). What intercultural rhetoric can bring to EAP/ESP writing studies in an English as a lingua Franca world. *Journal of English for Academic Purposes*, 29, 12–20. <https://doi.org/10.1016/j.jeap.2017.09.001>
- [McVee, M. B., Dunsmore, K., & Gavelek, J. R.](#) (2005). Schema theory revisited. *Review of Educational Research*, 75, 531–566.

- [Mitchell, R., Myles, F., & Marsden, E.](#) (2019). *Second language learning theories* (4th ed.). New York, NY: Routledge.
- [Mori, Y.](#) (1998). Effects of first language and phonological accessibility on Kanji recognition. *British Journal of Psychology*, 55, 165–180.
- [Nassaji, H.](#) (2002). Schema theory and knowledge-based processes in second language reading comprehension: A need for alternative perspectives. *Language Learning*, 52, 439–481.
- [Nation, I. S. P.](#) (2013). *Learning vocabulary in another language* (2nd ed.). Cambridge, England: Cambridge University Press.
- [Newton, J. M., Ferris, D. R., Goh, C. C. M., Grabe, W., Stoller, F. L., & Vandergrift, L.](#) (2018). *Teaching English to second language learners in academic contexts: Reading, writing, listening, and speaking*. New York, NY: Routledge.
- [Olive, T.](#) (2012). Writing and working memory: A summary of theories and findings. In E. L. Grigorenko, E. Mambrino, & D. D. Preiss (Eds.), *Handbook of writing: A mosaic of perspectives and views* (pp. 125–140). New York, NY: Psychology Press.
- [Olson, D. R.](#) (2012). Language, literacy, and mind: The literacy hypothesis. In E. L. Grigorenko, E. Mambrino, & D. D. Preiss (Eds.), *Handbook of writing: A mosaic of perspectives and views* (pp. 37–43). New York, NY: Psychology Press. (Reprinted from *Psychke*, 18, 2009).
- [Panetta, C. G.](#) (2001). *Contrastive rhetoric revisited and redefined*. Mahwah, NJ: Lawrence Erlbaum.
- [Powell, B. B.](#) (2012). *Writing: Theory and history of the technology of civilization*. Malden, MA: Wiley-Blackwell.
- Reichelt, M. (2016). L2 writers in languages other than English. In R. Manchón & P. K. Matsuda (Eds.), *Handbook of second and foreign language writing* (pp. 181–200). Berlin, Germany: de Gruyter Mouton.
- [Riazi, M., Shi, L., & Haggerty, J.](#) (2018). Analysis of the empirical research in the *Journal of Second Language Writing* at its 25th year (1992–2016). *Journal of Second Language Writing*, 41, 41–54.
- [Richards, T. L., Berninger, V. W., & Fayol, M.](#) (2012). The writing brain of normal child writers and children with writing disabilities: Generating ideas and transcribing them through the orthographic loop. In E. L. Grigorenko, E. Mambrino, & D. D. Preiss (Eds.), *Handbook of writing: A mosaic of perspectives and views* (pp. 85–105). New York, NY: Psychology Press.
- Roca de Larios, J., Nicolás-Conesa, F., & Coyle, Y. (2016). Focus on writers: Processes and strategies. In R. Manchón & P. K. Matsuda (Eds.), *Handbook of second and foreign language writing* (pp. 267–286). Berlin, Germany: de Gruyter Mouton.
- [Rogers, H.](#) (2004). *Writing systems: A linguistic approach*. Malden, MA: Wiley Blackwell.
- [Rumelhart, D. E.](#) (1980). Schemata: The building blocks of language. In R. J. Spiro, B. Bruce, & W. Brewer (Eds.), *Theoretical issues on reading comprehension* (pp. 33–58). Hillsdale, NJ: Lawrence Erlbaum.
- [Rumelhart, D. E., & McClelland, J. L.](#) (1982). An interactive activation model of the effects of context in perception. *Psychological Review*, 89, 60–94.
- [Rumelhart, D. E., Smolensky, P., McClelland, J. L., & Hinton, G. E.](#) (1986). Schemata and sequential thought processes in PDP models. In J. L. McClelland, D. E. Rumelhart, & the

- PDP Research Group (Eds.), *Parallel distributed processing: Explorations in the micro-structure of cognition—Psychological and biological models* (Vol. 2, pp. 7–57). Cambridge, MA: MIT Press.
- [Ryan, D.](#) (2020). Linguists' descriptions of the English writing system. In V. Cook & D. Ryan (Eds.), *Routledge handbook of the English writing system* (pp. 41–64). New York, NY: Routledge.
- Sasaki, M. (2016). L2 writers in study-abroad contexts. In R. Manchón & P. K. Matsuda (Eds.), *Handbook of second and foreign language writing* (pp. 161–180). Berlin, Germany: de Gruyter Mouton.
- [Schiffrin, D.](#) (1994). *Approaches to discourse*. Oxford, England: Blackwell.
- Schleppegrell, M., & Christie, F. (2018). Linguistic features of writing development: A functional perspective. In C. Bazerman, A. N. Applebee, V. W. Berninger, D. Brandt, S. Graham, J. V. Jeffery, P. K. Matsuda, S. Murphy, D. W. Rowe, M. Schleppegrell, & K. C. Wilcox (Eds.), *The lifespan development of writing* (pp. 111–150). Urbana, IL: National Council of Teachers of English.
- [Schmandt-Besserat, D.](#) (2012). Tokens as precursors of writing. In E. L. Grigorenko, E. Mambrino, & D. D. Preiss (Eds.), *Handbook of writing: A mosaic of perspectives and views* (pp. 3–10). New York, NY: Psychology Press.
- [Seidenberg, M.](#) (2017a). *Reading at the speed of sight*. New York, NY: Basic Books.
- [Seidenberg, M.](#) (2017b). *Reading: Our new understanding of the original information technology*. New York, NY: Basic Books.
- [Seloni, L., & Lee, S. H.](#) (Eds.). (2019). *Second language writing instruction in global contexts: English language teacher preparation and development*. Bristol, England: Multilingual Matters.
- [Shortis, T.](#) (2020). Texting and other messaging: Written system in digitally mediated vernaculars. In V. Cook & D. Ryan (Eds.), *Routledge handbook of the English writing system* (pp. 487–516). New York, NY: Routledge.
- [Smith, F.](#) (2004). *Understanding reading* (6th ed.). Mahwah, NJ: Lawrence Erlbaum.
- [Snow, M. A.](#) (2005). Key themes in TESOL MA teacher education. In D. Tedick (Ed.), *Second language teacher education: International perspectives* (pp. 261–272). Mahwah, NJ: Lawrence Erlbaum.
- [Sproat, R.](#) (2020). English among the writing systems of the world. In V. Cook & D. Ryan (Eds.), *Routledge handbook of the English writing system* (pp. 27–40). New York, NY: Routledge.
- [Squires, L.](#) (2020). Computer-mediated communication and the English writing system. In V. Cook & D. Ryan (Eds.), *Routledge handbook of the English writing system* (pp. 471–486). New York, NY: Routledge.
- [Tardy, C. M.](#) (2017). The challenge of genre in the academic writing classroom: Implications for L2 writing teacher education. In J. Bitchener, N. Storch, & R. Wette (Eds.), *Teaching writing for academic purposes to multilingual students: Instructional approaches* (pp. 69–83). New York, NY: Routledge.
- [Tardy, C. M., & Whittig, E.](#) (2017). On the ethical treatment of EAL writers: An update. *TESOL Quarterly*, 51(4), 920–930.

- [Thatcher, B.](#) (2011). *Intercultural rhetoric and professional communication: Technological advances and organizational behavior*. Hershey, PA: IGI Global.
- [Thatcher, B., & St. Amant, K.](#) (Eds.). (2010). *Teaching intercultural rhetoric and technical communication: Theories, curriculum, pedagogies, and practices*. Amityville, NY: Baywood.
- [Toledo, P. F.](#) (2005). Genre analysis and reading of English as a foreign language: Genre schemata beyond text typologies. *Journal of Pragmatics*, 37, 1059–1079.
- [Tracey, D. H., & Morrow, L. M.](#) (2017). *Lenses on reading: An introduction to theories and models* (3rd ed.). New York, NY: Guilford.
- [Tsu, J.](#) (2022). *Kingdom of characters: The language revolution that made China modern*. New York, NY: Penguin.
- [Tsui, A. B. M.](#) (2009). Teaching expertise: Approaches, perspectives, and characterizations. In A. Burns & J. C. Richards (Eds.), *The Cambridge guide to second language teacher education* (pp. 190–198). Cambridge, England: Cambridge University Press.
- [Venezky, R. L.](#) (1995). How English is read: Grapheme-phoneme regularity and orthographic structure in word recognition. In I. Taylor & D. R. Olson (Eds.), *Scripts and literacy: Reading and learning to read alphabets, syllabaries, and characters* (pp. 111–130). Dordrecht, The Netherlands: Kluwer Academic.
- [Willingham, D. T.](#) (2017). *The reading mind: A cognitive approach to understanding how the mind reads*. San Francisco, CA: Jossey-Bass.
- [Wolf, M.](#) (2007). *Proust and the squid: The story and science of the reading brain*. New York, NY: HarperCollins.
- [Wolf, M.](#) (2016). *Tales of literacy for the 21st century*. Oxford, England: Oxford University Press.
- [Wolf, M.](#) (2018). *Reader, come home: The reading brain in a digital world*. New York, NY: HarperCollins.
- [Yap, M. J., & Liow, S. J. R.](#) (2020). Processing the written word. In V. Cook & D. Ryan (Eds.), *Routledge handbook of the English writing system* (pp. 453–469). New York, NY: Routledge.
- [You, X.](#) (2010). *Writing in the devil's tongue: A history of English composition in China*. Carbondale, IL: Southern Illinois University Press.
- [You, X.](#) (Ed.). (2018). *Transnational writing education: Theory, history, and practice*. New York, NY: Routledge.
- [Yule, V., & Yasuko, I.](#) (2020). Spelling reform. In V. Cook & D. Ryan (Eds.), *Routledge handbook of the English writing system* (pp. 413–427). New York, NY: Routledge.
- [Zheng, B., & Warschauer, M.](#) (2017). Epilogue: Second language writing in the age of computer-mediated communication. *Journal of Second Language Writing*, 36, 61–67. <http://dx.doi.org/10.1016/j.jslw.2017.05.014>
- [Zhu, W.](#) (2010). Theory and practice in second language writing: How and where do they meet? In T. Silva & P. K. Matsuda (Eds.), *Practicing theory in second language writing* (pp. 209–228). West Lafayette, IN: Parlor Press.

[Zins, J. M., & Hooper, S. R.](#) (2012). The interrelationship of child development and written language development. In E. L. Grigorenko, E. Mambrino, & D. D. Preiss (Eds.), *Handbook of writing: A mosaic of perspectives and views* (pp. 47–64). New York, NY: Psychology Press.